Problems with Chinchilla Approach 2: Systematic Biases in IsoFLOP Parabola Fits

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Abstract

Chinchilla Approach 2 is among the most widely used methods for fitting neural scaling laws. Its parabolic approximation introduces systematic biases in compute-optimal allocation estimates, even on noise-free synthetic data. Applied to published Llama 3 IsoFLOP data at open frontier compute scales, these biases imply a parameter underallocation corresponding to 6.5% of the 3.8×10^{25} FLOP training budget and \$1.4M (90% CI: \$412K–\$2.9M) in unnecessary compute at 50% H100 MFU. Simulated multimodal model misallocations show even greater opportunity costs due to higher loss surface asymmetry. Three sources of this error are examined: IsoFLOP sampling grid width (Taylor approximation accuracy), uncentered IsoFLOP sampling, and loss surface asymmetry ($\alpha \neq \beta$). Chinchilla Approach 3 largely eliminates these biases but is often regarded as less data-efficient, numerically unstable, prone to local minima, and harder to implement. Each concern is shown to be unfounded or addressable, especially when the partially linear structure of the objective is exploited via Variable Projection, enabling unbiased inference on all five loss surface parameters through a two-dimensional optimization that is well-conditioned, analytically differentiable, and amenable to dense, or even exhaustive, grid search. It may serve as a more convenient replacement for Approach 2 or a more scalable alternative for adaptations of Approach 3 to richer scaling law formulations. See https://github.com/Open-Athena/vpnls for details and https://openathena.ai/scaling-law-analysis for other results from this study.

1 Introduction

The Chinchilla paper [Hoffmann et al., 2022] introduced three approaches to scaling law estimation, and of these, Approach 2 has arguably seen the broadest adoption. It has been used by leading AI labs including DeepMind [Hoffmann et al., 2022, Alabdulmohsin et al., 2023] (its creators), Meta [Grattafiori et al., 2024, Dervishi et al., 2025, Tong et al., 2026], DeepSeek [DeepSeek-AI et al., 2024], Microsoft [Zhang et al., 2025], Amazon [Tuyls et al., 2024], Waymo [Baniodeh et al., 2025], and Arc Institute [Nguyen et al., 2024], among others (see Table A2). It is also a workhorse method for academic scaling law studies [Liang et al., 2024, von Rütte et al., 2025, Fortin, 2024] and tutorials[1] like those from Andrej Karpathy.

The method's appeal lies in its putative stability and data efficiency relative to nonlinear optimization over all loss surface parameters. Rather than fitting all five parameters of the loss surface simultaneously, Approach 2 targets only the two scaling exponents, relying on second-order Taylor approximations that reduce each IsoFLOP curve to a simple parabola. This sacrifices recovery of the full loss surface but makes estimation simple, and enables extraction of what are typically

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[1] https://github.com/karpathy/nanochat/discussions/420

viewed as the most actionable scaling properties of a model and dataset. It does this through a sequence of straightforward quadratic and linear fits, which frequently requires far more than 5 parameters to be estimated (at least 3 per compute budget and 2 for the final power law), but elides the need for a nonlinear optimizer.

Despite this broad adoption, the sensitivity of the method's core approximations and its behavior on loss surfaces that are less symmetric than the original Chinchilla form (where parameter and token scaling exponents are roughly equal) have not, to our knowledge, been studied in detail. Here we revisit the basics of how to fit the Chinchilla loss model with high precision and stability. We investigate this through 1) noise-free synthetic simulations of IsoFLOP experiments, 2) closed-form expression of one particular Approach 2 error mode, 3) simulations with a noise model justified by examining residuals from 6 real IsoFLOP experiments, each filtered through an extensive 8-step quality control pipeline, and 4) comparing allocation forecasts from Approach 2 to other methods after fitting them on Llama 3 training runs.

We also show how extrapolation errors trace back to suboptimal IsoFLOP experiment design, and that pathologies in these designs can be observed in real, high-profile scaling law studies even if they are difficult to quantify precisely. Finally, we propose an alternative fitting method that is simple, stable, and free of these biases while building on the same computational shortcut as Approach 2: optimizing exponential terms separately from linear terms. We call this approach Variable Projection with Non-negative Least Squares (VPNLS).

This investigation is also motivated by a broader landscape of *analytical* extensions to the Chinchilla loss surface. A growing body of work adds or modifies terms in the original functional form to account for additional training configuration choices such as data repetition [Muennighoff et al., 2023], overfitting [Qu et al., 2024], precision [Kumar et al., 2024], optimizers [Volkova et al., 2026], MoE sparsity [Ludziejewski et al., 2025], pruning [Jin et al., 2025], data quality [Subramanyam et al., 2025], data mixtures [Shukor et al., 2025a, Longpre et al., 2025], and downstream task performance [Bhagia et al., 2024], to name a few. These studies prescribe explicit functional forms rather than inferring scaling law structure automatically. Related work analyzes the effects of width, depth, and context length in a theoretical in-context regression model [Bordelon et al., 2025], and the effect of counting non-embedding parameters on fitted allocation exponents [Pearce and Song, 2024]. Similar studies [Goyal et al., 2024, Tian et al., 2025, Rolf et al., 2021] extend individual terms in isolation (e.g. token scaling terms alone) or propose modified Kaplan scaling laws [Kaplan et al., 2020], both of which would be relevant for comparable Chinchilla formulations as well. A fitting method that recovers the base surface with greater scalability and stability may therefore offer a stronger starting point for these adaptations.

2 Preliminaries

Neural scaling laws describe how model performance improves with compute. The Chinchilla loss surface models this relationship as:

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}} \tag{1}$$

where N is the number of parameters, D is the number of training tokens, E is the irreducible loss, and A, B, α, β capture how quickly performance improves with scale.

Given a compute budget $C \approx 6ND$, the optimal allocation satisfies:

$$N^* \propto C^a \quad \text{where} \quad a = \frac{\beta}{\alpha + \beta} \tag{2}$$

$$D^* \propto C^b \quad \text{where} \quad b = \frac{\alpha}{\alpha + \beta} \tag{3}$$

Recovering just the exponents a and b from empirical training runs is useful for a variety of purposes. Two canonical approaches for this exist:

2.1 Approach 2: IsoFLOP Parabolic Fitting

This approach takes advantage of the fact that along a fixed-compute contour (IsoFLOP curve), loss as a function of $\log N$ is approximately parabolic near the optimum. The procedure has three steps:

1. **Sample IsoFLOP contours:** For each compute budget C, train models at various (N, D) pairs satisfying $C = 6ND$.
2. **Fit parabolas:** For each budget, fit $L = p(\log N)^2 + q(\log N) + r$ and extract the minimum N^*.
3. **Fit power laws:** Regress $\log N^*$ against $\log C$ to recover the exponent a (and similarly for D^*, b).

The appeal is simplicity: only polynomial fits, no nonlinear optimization. The parabolic approximation comes from a Taylor expansion of the loss surface around the optimum.

2.2 Approach 3: Direct Surface Fitting

The alternative is to fit all five parameters (E, A, B, α, β) simultaneously by minimizing the residual sum of squares (RSS) directly:

$$\min_{E,A,B,\alpha,\beta} \sum_i \big(L_i - \hat{L}(N_i, D_i)\big)^2 \quad \text{where} \quad \hat{L}(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta} \tag{4}$$

At least two practical issues arise with this direct formulation. First, E, A, and B must remain positive for the surface to be physically meaningful, requiring box constraints. Second, the prediction involves a sum of exponentials that can span many orders of magnitude, creating numerical optimization difficulties.

The Chinchilla paper [Hoffmann et al., 2022] addresses both issues with an adapted formulation. A LogSumExp (LSE) reparameterization optimizes (e, a, b, α, β) where $E = e^e$, $A = e^a$, $B = e^b$, enforcing positivity without explicit bounds. The predicted log-loss is computed as:

$$\log \hat{L}(N, D) = \text{logsumexp}\big(e,\ a - \alpha \log N,\ b - \beta \log D\big) \tag{5}$$

The objective is then a Huber loss on log-predictions rather than MSE on predictions. Operating in log-space improves numerical conditioning, and the Huber function provides robustness to outliers from noisy training runs.

Because we evaluate methods on simulated data with known noise models and want maximum likelihood estimates in some configurations, we use MSE rather than the Huber objective throughout this study. This is the maximum likelihood estimator (MLE) under the constant, additive Gaussian

noise assumed in our simulations. When fitting real data, we remove outliers before fitting in most cases. Figures A8 and A9 examine residual variance across compute budgets in six real IsoFLOP experiments and find no strong evidence that noise varies by budget, supporting this assumption. Some more details on these differences in optimization objectives, reparameterizations, outcome scaling, and how they relate to common implementations are discussed in Appendix A and the exponent inference comparison (Section 5.4).

3 Error Costs: Misallocation at Scale

Scaling laws serve at least two broad purposes in practice. At smaller scales they guide research decisions like comparing architectures, datasets, and training strategies before committing to a full run. For production models of small to medium size, allocation planning typically targets deliberate overtraining (training on more data than compute-optimal) rather than compute-optimality itself, and Approach 3 or variations on it are much more common in that setting. The compute-optimal scaling laws analyzed in this work are most directly relevant to the largest models in frontier families, which are still frequently trained near compute-optimality. This means that the Approach 2 biases detailed in later sections are applicable, and may translate directly into wasted compute.

To quantify this, we assess the cost of Approach 2 misallocations at the compute scale of Llama 3 405B (3.8×10^{25} FLOPs) [Grattafiori et al., 2024]. Epoch AI used this model as a high-compute open-model reference point in its January 2025 analysis [Emberson, 2025]. We also examine three additional loss surfaces with published Approach 3 fit statistics—the original Chinchilla surface [Hoffmann et al., 2022], and two multimodal models for which scaling is substantially more asymmetric, SODA [Manakul et al., 2026] (interleaved audio-text) and Sparse-NMM [Shukor et al., 2025b] (native multimodal with sparse mixture-of-experts). These surfaces span a range of exponent asymmetry ratios (α/β) from near-unity to nearly 2:1, providing a view of how misallocation grows as the loss surface departs from nearly equal scaling exponents.

We measure misallocation in terms of **Deadweight Compute Loss** (DCL), defined as the excess FLOPs consumed by training at the Approach 2–inferred allocation relative to the budget that would achieve the same loss under optimal allocation. Given a compute budget C and an inferred data allocation D^*, the model size is forced by the constraint $N = C/(6D^*)$. The resulting loss $L(N, D^*)$ is suboptimal. Inverting the optimal loss function $L_{\text{opt}}(C)$ yields the smaller budget C_{eq} that would reach this same loss with optimal allocation; $\text{DCL} = C - C_{\text{eq}}$. This is analogous to deadweight loss in economics, where dataset and model sizes define a supply–demand tradeoff that maximizes a "total surplus"—or equivalently, minimizes validation loss—only at the compute-optimal allocation. Any departure from optimality wastes compute in the same sense that a price distortion wastes economic surplus.

For Llama 3, we digitize individual IsoFLOP data points[2] from Figure 2 of the original paper [Grattafiori et al., 2024] following the SVG-coordinate extraction method of Besiroglu et al. [Besiroglu et al., 2024], then fit both Approach 2 and two Approach 3 implementations (VPNLS and direct L-BFGS-B) to obtain competing allocations at the target budget. For Chinchilla, SODA, and Sparse-NMM we take a simulated approach instead, generating synthetic IsoFLOP data from each published loss surface, fitting Approach 2 to it under controlled bias conditions (narrow sampling grid, drifting sampling centers) detailed in later sections, and comparing the resulting extrapolations. We include a simulated Llama 3 condition as well, which provides an implicit measure of how much of the empirical Llama 3 error is attributable to the real IsoFLOP experiment design versus

[2] https://github.com/eric-czech/llama3_isoflop_extraction

the biases we introduce systematically. Dollar costs are computed assuming 50% model FLOPs utilization [Sardana et al., 2024] and $2 per H100-hour [Team Olmo et al., 2025].

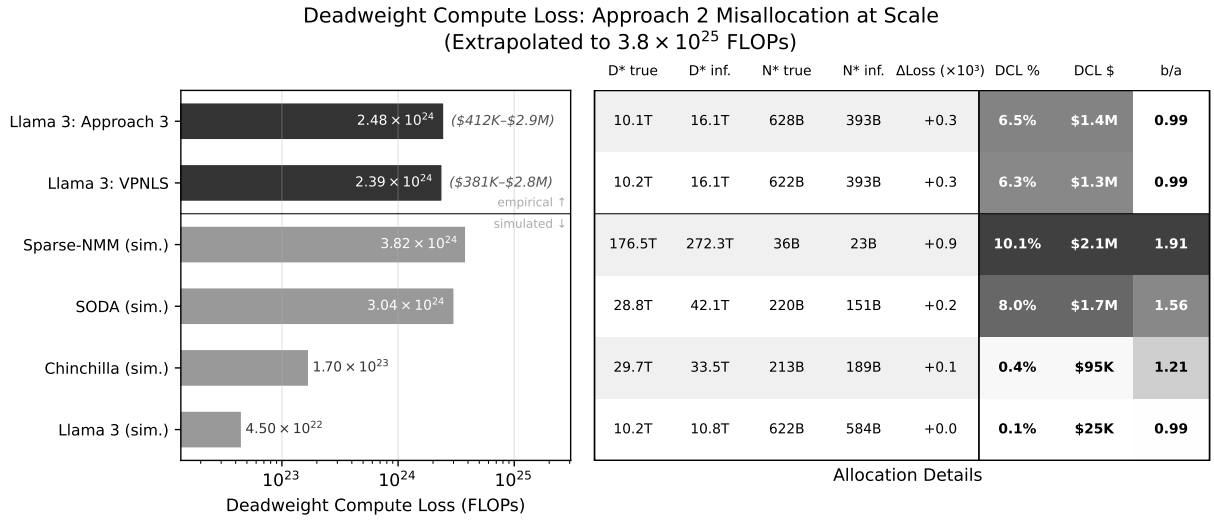


Figure 1: Approach 2 misallocation costs extrapolated to 3.8×10^{25} FLOPs. Left: Deadweight Compute Loss (DCL) as a percentage of budget; dollar cost ranges for empirical rows are 90% bootstrap CIs. Right: allocation details including true vs. inferred token counts and model sizes, loss penalty, and dollar cost. "Empirical" rows use Approach 2 power laws fit to digitized Llama 3 IsoFLOP data, evaluated against VPNLS and Approach 3 surfaces. "Simulated" rows use synthetic IsoFLOP data generated from each published surface with the XS grid and 3× drift bias used in the Drifting Bias simulation section (Section 4.3.2).

Figure 1 shows the results. On the empirical Llama 3 data, Approach 2 overestimates the compute-optimal token count by ∼60% relative to the Approach 3 surface, yielding DCL of 6.5% of budget ($1.4M). Notably, our Approach 2 fit yields $b = 0.5368$, which rounds exactly to 0.537 as reported in the Llama 3 paper. These results also include extrapolations from the Variable Projection (VPNLS) method discussed in Section 5.2. This provides a contrast between projections made from a canonical Approach 3 implementation fit on logarithmic loss values and one without this rescaling, a necessary assumption for unbiased estimation in our later simulations (see Appendix A). The simulated Llama 3 condition, which isolates the systematic biases from the IsoFLOP experiment design, produces only 0.2% DCL ($39K) at $\beta/\alpha \approx 0.97$. This suggests that the magnitude of the off-center sampling bias assumed, which matches the most extreme form of that bias used in later sections, has little impact relative to the biases present in the actual Llama 3 IsoFLOP data (see Figure 8).

Figure 2 shows that when known Approach 2 biases are addressed through targeted quality control (QC), Approach 2 and Approach 3 converge on nearly identical estimates with negligible deadweight loss. Figure A1 visualizes that QC pipeline across six experiments spanning different datasets and model families, including Chinchilla, Llama 3, CoMMA, DCLM, Nemotron, and FineWeb. Figure A7 shows the same analysis for Chinchilla data, extrapolated to its own evaluation budget of 5.8×10^{23} FLOPs. DCL drops from 37.9% to 12.5%, demonstrating a similar convergence in allocation estimates.

While the simulated Llama 3 surface produces minimal waste due to its nearly symmetric exponents, surfaces with greater asymmetry show substantially higher costs. SODA ($\alpha/\beta \approx 1.56$) reaches 8.0% DCL ($1.7M) and Sparse-NMM ($\alpha/\beta \approx 1.91$) reaches 10.1% ($2.1M). For domains with highly asymmetric scaling exponents, which can also occur in multimodal settings [Tong et al.,

2026], Approach 2 misallocations can waste a non-trivial fraction of a large training budget.

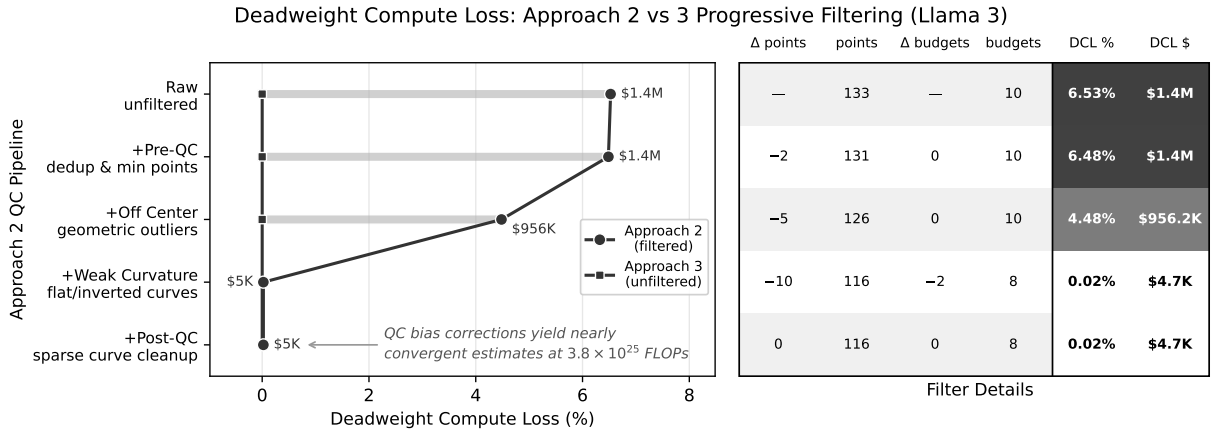


Figure 2: Effect of progressive quality control filtering on Approach 2 Deadweight Compute Loss, measured against an Approach 3 surface fit to unfiltered Llama 3 data. Each row cumulatively applies one QC stage. Nearly all DCL reduction comes from the off-center and weak-curvature filters, which target specific Approach 2 biases. The right table reports points and budgets removed at each stage, along with DCL as a percentage of the 3.8×10^{25} FLOP evaluation budget and dollar cost.

The source of these Approach 2 misallocations is characterized in many of the following sections. This begins with simulations of IsoFLOP experiments under perfect, unrealistic conditions and progresses to noisier, more realistic scenarios. Each of these demonstrates one or more estimation biases that can compound significantly in extrapolations, as we attempted to show more directly in this section.

4 Error Sources: Approach 2 Biases

4.1 Symmetric Surfaces

Before examining failure modes, we establish that Approach 2 works perfectly under ideal conditions. Consider a **symmetric** loss surface where $\alpha = \beta$:

$$L(N, D) = 1.69 + \frac{400}{N^{0.31}} + \frac{400}{D^{0.31}} \tag{6}$$

With equal exponents, the optimal allocation splits compute evenly between parameters and data. The true scaling exponents are:

$$a = b = \frac{0.31}{0.31 + 0.31} = 0.5 \tag{7}$$

We sample five IsoFLOP contours spanning 10^{17} to 10^{21} FLOPs, with 15 model sizes per curve, fit parabolas to each, and extract the optimal token count D^*. Most simulations in this study use these same five compute budgets and 15 points per IsoFLOP curve, unless stated otherwise.

The results confirm perfect recovery of the token scaling exponent and intercept as shown in Figure 3 and Table 1. This establishes our baseline. Approach 2 is precisely correct under ideal conditions that are unrealistic in practice.

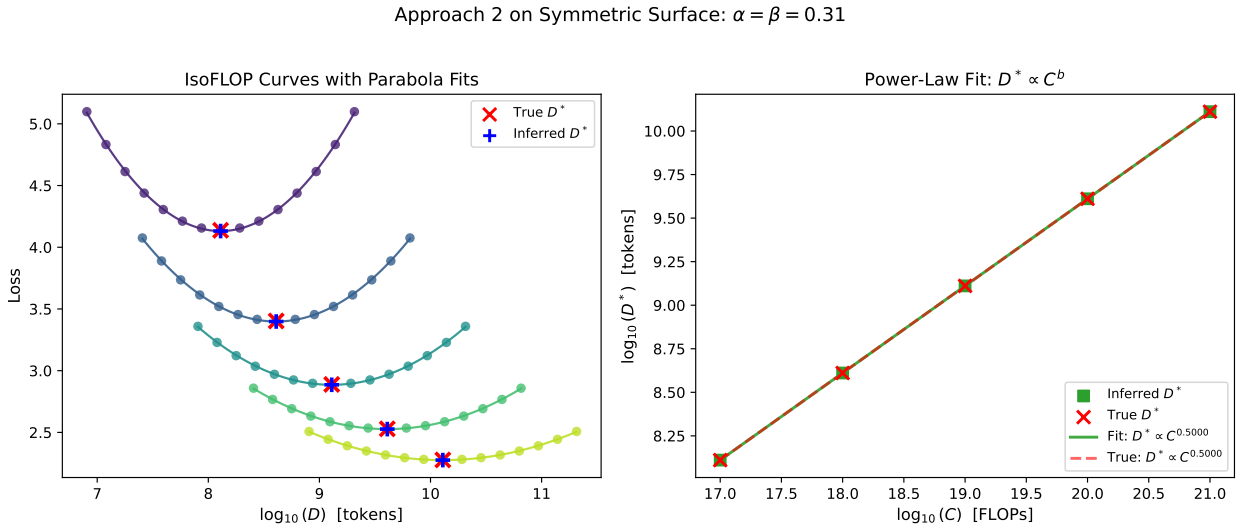


Figure 3: Approach 2 applied to a symmetric loss surface. Left: IsoFLOP curves with fitted parabolas. True (×) and inferred (+) optima are indistinguishable. Right: Power-law fit recovers the exact scaling exponent.

Parameter	True Value	Inferred Value	Relative Error
b (D^* exponent)	0.500000	0.500000	$+6.2 \times 10^{-12}\%$
b_0 (D^* intercept)	-0.389076	-0.389076	$-1.4 \times 10^{-10}\%$

Table 1: Approach 2 parameter recovery on the symmetric surface.

4.2 Asymmetric Surfaces

We repeat the exact same procedure as before: perfect sampling centers, no noise, identical methodology. The only change is that the loss surface is now **asymmetric** ($\alpha \neq \beta$). Simulation results for this scenario show that Approach 2 produces systematically wrong intercepts while exponents remain accurate. This means that fitting Chinchilla Approach 2 to noise-free data drawn from the published Chinchilla loss surface produces an incorrect scaling law, a result we found to be counterintuitive.

We test two configurations to see how the effect scales:

- **Chinchilla:** $\alpha = 0.34$, $\beta = 0.28$ (ratio ≈ 1.2)
- **Asymmetric:** $\alpha = 0.46$, $\beta = 0.15$ (ratio $= 3.0$)

The Asymmetric surface is an illustrative stress case with $\alpha/\beta = 3$. Table A2 includes empirical allocation ratios in the 2–3 range and higher [Cheng et al., 2024], although these ratios do not by themselves identify loss-surface asymmetry. Modality-dependent allocation patterns also occur in multimodal work [Tong et al., 2026]. Our simulations isolate the effect of loss-surface imbalance; its magnitude, rather than its direction, drives most of the biases studied here.

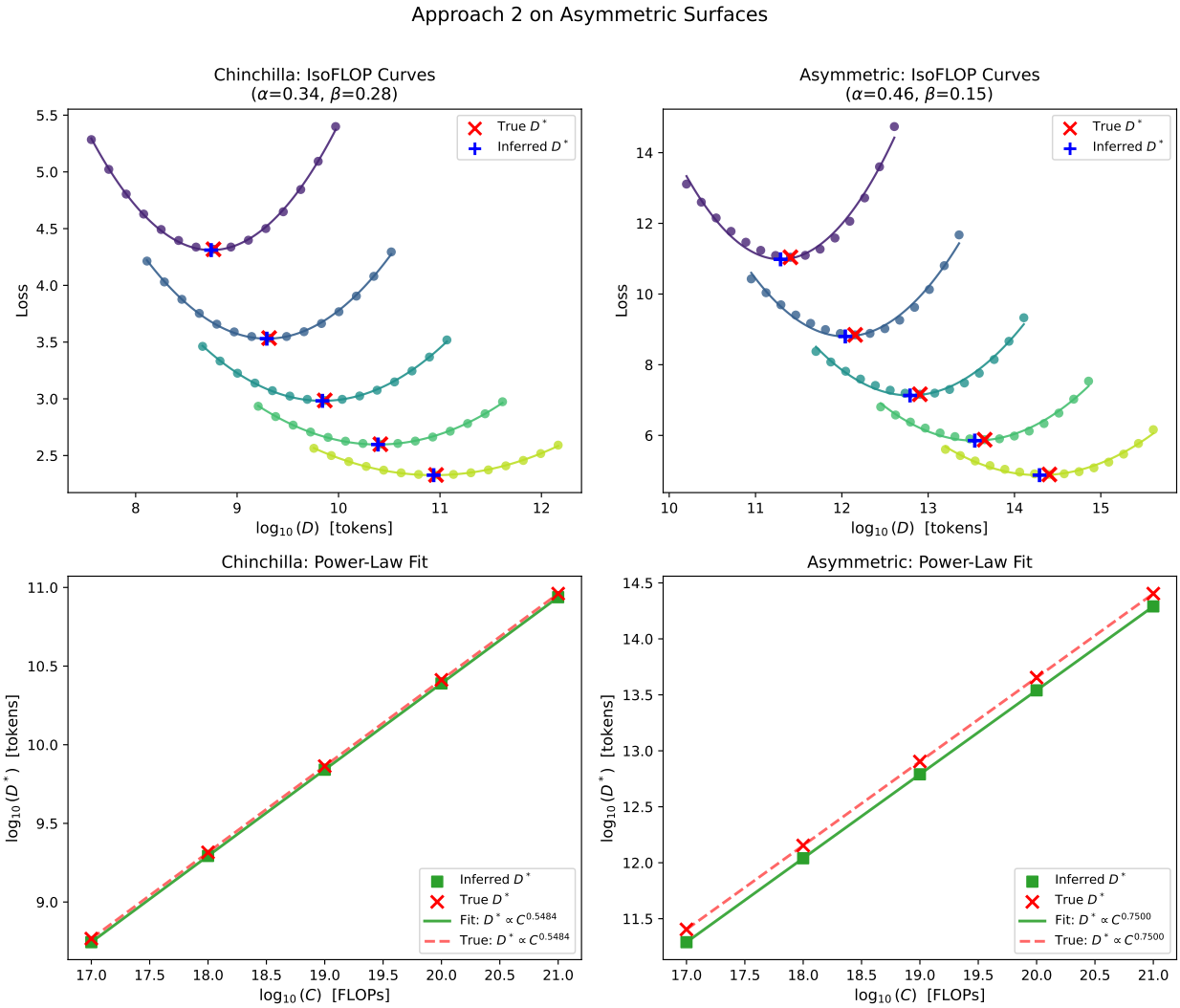


Figure 4: Approach 2 on asymmetric loss surfaces. Note the visible gap between true (dashed) and inferred (solid) power-law lines in the Asymmetric case. The exponents match perfectly, but the intercepts differ.

Surface	Parameter	True Value	Inferred Value	Relative Error
Chinchilla	b (D^* exponent)	0.548387	0.548387	$\approx 0\%$
	b_0 (D^* intercept)	-0.555357	-0.578092	-4.1%
Asymmetric	b (D^* exponent)	0.750000	0.750000	$\approx 0\%$
	b_0 (D^* intercept)	-1.345791	-1.459957	-8.5%

Table 2: Approach 2 parameter recovery on asymmetric surfaces.

Figure 4 shows the results from this simulation, with relative errors in Table 2. Both show how errors increase with greater asymmetry and the following section outlines how they arise.

4.2.1 Underlying Causes

The IsoFLOP loss curve is not a true parabola. When a parabola is fit to this curve, the parabola's minimum (vertex) does not land exactly at the true optimum. It shifts slightly, and this shift depends only on the loss surface shape (α, β) and the sampling grid. It does not depend on compute budget. The sampling grid size becomes important here as wider grids amplify the mismatch between the true curve and its parabolic approximation, increasing the vertex shift.

Because the IsoFLOP parabola is fit in $\log N$ space (as described in the Approach 2 procedure), the vertex shift directly biases N^*. Since $C = 6ND$, analyzing the bias in either N^* or D^* is sufficient, and we focus on N^* below.

Since the vertex shift is constant across all compute budgets, it biases every inferred N^* by the same multiplicative factor. When fitting $\log N^*$ vs $\log C$ to extract scaling exponents:

- The **slope (exponent)** is unchanged: multiplying all N^* values by a constant factor adds a constant to $\log N^*$, which does not affect the slope.
- The **intercept** absorbs the entire error, biased by exactly that multiplicative factor.

The intercept error can be derived analytically in closed form. The parabola vertex shifts by δw (in log-space), giving an intercept error of:

$$\text{Intercept error} = 10^{\delta w} - 1 \tag{8}$$

where $\delta w = f(\alpha, \beta, W, n)$ depends only on the surface exponents and the sampling grid (width W in log-space, number of points n per IsoFLOP curve), not on C, E, A, or B. Here W spans $10^{-W/2}$ to $10^{W/2}$ times the optimal N^*, so $W = 2.41$ (a $\pm 16\times$ grid) means sampling from $\frac{1}{16}\times$ to $16\times$ the optimum. Key properties:

- $\delta w = 0$ when $\alpha = \beta$ (symmetric surfaces have no error)
- δw grows with $|\alpha - \beta|$ (more asymmetry, more error)
- δw grows with W (wider sampling range, more error)

For example, with the Chinchilla parameters ($\alpha = 0.34$, $\beta = 0.28$): a $\pm 2\times$ grid ($W = 0.60$) yields 0.3% intercept error, while a $\pm 16\times$ grid ($W = 2.41$) yields 4.1% error.

The full derivation[3] provides the closed-form expression for vertex shift δw as a function of α, β, W, and n. It also shows how this shift translates directly into intercept error, independent of compute budget.

Intuition via Taylor expansion. A parabola is a 2nd-order polynomial, equivalent to a 2nd-order Taylor expansion around the optimum. The approximation $L(w) \approx L(0) + \frac{1}{2}L''(0)w^2$ is only valid when higher-order terms are negligible, i.e., when samples are close to the true minimum. As sampling range increases, 3rd and 4th order terms grow. For symmetric surfaces ($\alpha = \beta$), odd-order terms cancel by symmetry, preserving the vertex location. For asymmetric surfaces, they do not cancel, shifting the fitted vertex away from the true optimum.

[3] https://github.com/Open-Athena/scaling-law-analysis/blob/main/results/article/static/scaling_parameter_errors.md

4.2.2 Error Implications

Extrapolation to higher compute budgets requires both exponents and intercepts to be correct. The previous subsection (Section 4.2.1) established that asymmetric loss surfaces produce provably biased intercepts even under ideal experimental conditions. Here we quantify what those errors mean in practical terms by examining compute-optimal token prediction: given a compute budget, how many tokens does the inferred scaling law predict?

Up to this point, all analysis has assumed a single fixed sampling grid width. We now examine how token prediction error varies with both compute budget and sampling grid width. For surfaces with asymmetric exponents, wider sampling grids amplify the parabola-fitting mismatch, increasing the constant vertex shift and thus the intercept bias. To make this comparison concrete, we first define what "wider" and "narrower" mean in quantitative terms.

A sampling grid of "$\pm k\times$" means the sampled values (whether model sizes or token counts) range from $\frac{1}{k}$ to k times the true optimum at each compute budget. The total range covered is k^2 (the ratio of largest to smallest), and the $\log_{10}$ of that ratio gives the number of decades the grid spans end-to-end. Table 3 shows the four grid widths used in this analysis.

Grid Name	$\pm k\times$	Sampling Range	Total Ratio	Decade Span
Extra Small (XS)	$\pm 2\times$	$\frac{1}{2}\times$ to $2\times$	$4\times$	0.60
Small (S)	$\pm 4\times$	$\frac{1}{4}\times$ to $4\times$	$16\times$	1.20
Large (L)	$\pm 8\times$	$\frac{1}{8}\times$ to $8\times$	$64\times$	1.81
Extra Large (XL)	$\pm 16\times$	$\frac{1}{16}\times$ to $16\times$	$256\times$	2.41

Table 3: Sampling grid widths used in simulations.

In practice, scaling law experiments typically sample across 1 to 2 decades in token count, placing the Small and Large grids squarely within the realistic range. The Extra Small and Extra Large grids bracket this range on either side, illustrating how the biases shrink or grow as the sampling window narrows or widens. The Extra Large grid ($\pm 16\times$, $\sim$2.4 decades) is the default used in all single-grid analyses in the preceding sections.

The key observations from Figure 5 are:

- **Symmetric surfaces are unaffected:** When $\alpha = \beta$, all grid widths produce zero error.
- **Asymmetric surfaces underestimate:** Negative errors mean the inferred D^* is smaller than the true D^*.
- **Wider grids amplify error:** Moving from XS ($\pm 2\times$) to XL ($\pm 16\times$) grids increases error from 0.3% to 5.1% on Chinchilla, and from 1.7% to 23% on the Asymmetric surface.
- **Asymmetry further amplifies error:** The Asymmetric surface ($\alpha/\beta = 3$) shows roughly 4–5$\times$ larger errors than Chinchilla at each grid width.

4.3 Off-Center Sampling

The previous simulation sections assumed perfectly centered sampling. At every compute budget, the IsoFLOP grid was placed exactly at the true optimum. In practice, N^* is not known before running the experiment. Sampling centers are guesses, informed by prior estimates or heuristics, and they will likely be wrong by some amount.

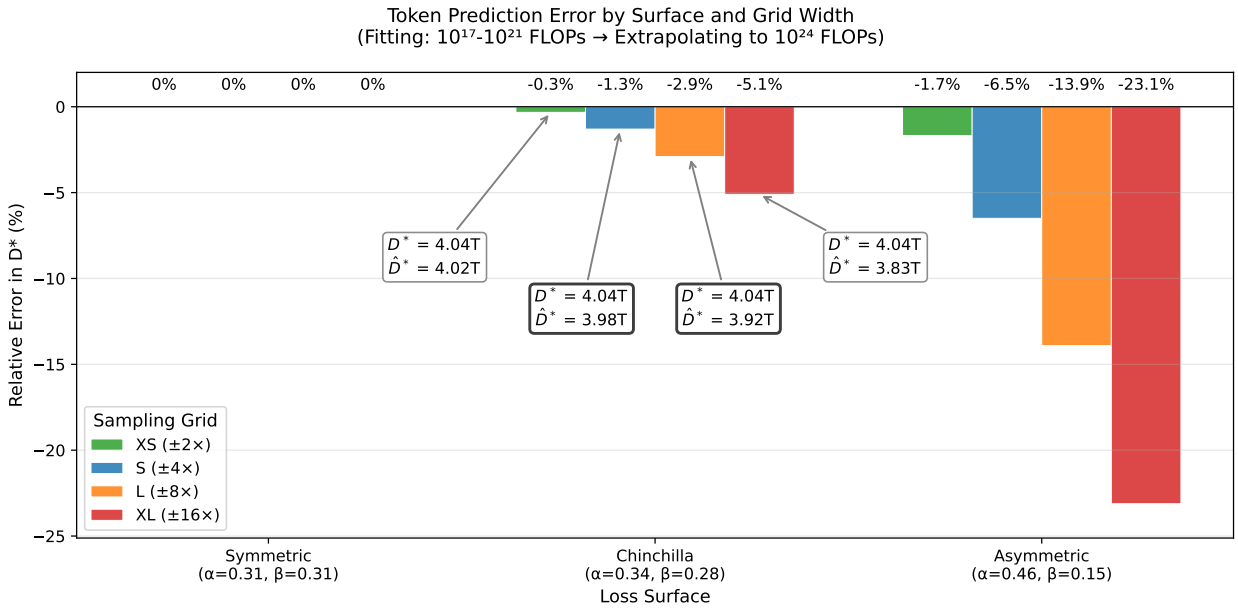


Figure 5: Relative error in compute-optimal token prediction when extrapolating from the training range (10^{17}–10^{21} FLOPs) to 10^{24} FLOPs. Negative values indicate underestimation: the inferred scaling law predicts fewer tokens than optimal. Bars are grouped by sampling grid width. Annotations for the Chinchilla surface show D^* (true compute-optimal token count) versus $\hat{D}^*$ (the Approach 2 estimate); the Small and Large grid annotations are emphasized (thicker borders) as they fall within the realistic 1–2 decade range typical of scaling law experiments, while Extra Small and Extra Large bracket either side as more extreme configurations.

This is a distinct source of error from the asymmetry bias examined earlier. Asymmetry errors arise from the shape of the loss surface ($\alpha \neq \beta$); off-center errors arise from where the sampling grid is placed. To isolate this new effect, we use the symmetric surface ($\alpha = \beta = 0.31$) again where asymmetry bias is zero by construction.

4.3.1 Constant Multiplicative Bias

The simplest form of off-center sampling is a constant multiplicative offset where every compute budget's sampling center is shifted by the same factor from the true optimum. A "3× offset" means each IsoFLOP grid is centered at $3 \times D^*$ instead of D^*, so the grid midpoint consistently sits at three times the true optimal token count.

Because this offset is the same at every compute budget, it has a familiar geometric effect where each parabola vertex shifts by a constant amount in log-space. This is the same mechanism as asymmetry bias. The slope of $\log D^*$ vs $\log C$ is unaffected (a constant additive shift in log-space does not change the slope), so the scaling exponent is preserved while the intercept is not.

The extrapolation in Figure 6 (top right) shows what this means for token prediction. All four grid widths overestimate D^*, with the narrowest grid (XS) producing the largest error. This is the reverse of the asymmetry bias pattern, where wider grids amplified error. Here, narrower grids are more sensitive to off-center placement because fewer samples lie near the true optimum.

The intercept error panel (bottom right) shows the same pattern across the full continuum of grid widths. The error is always positive (the inferred D^* overshoots) and decreases monotonically as the grid widens, reflecting how a wider sampling range brings more of the true loss curve's shape

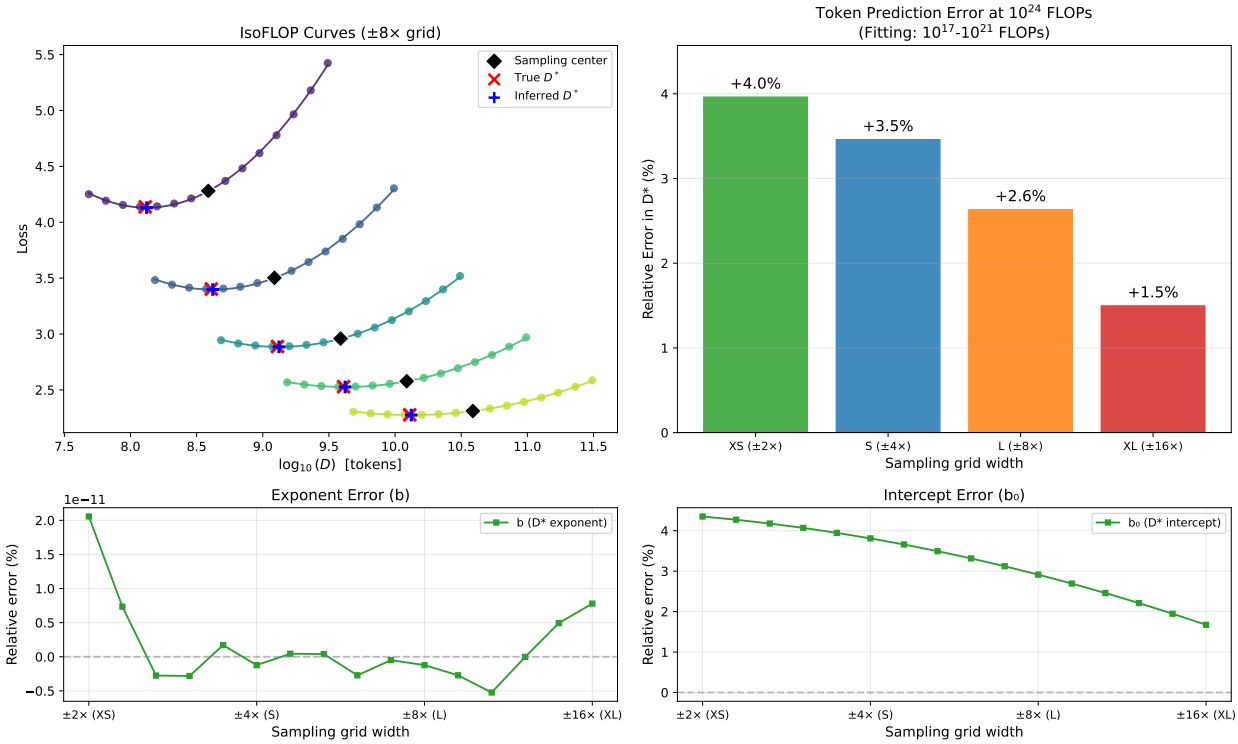


Figure 6: Effect of a constant 3× offset in sampling centers on the symmetric surface. Top left: IsoFLOP curves at the Large grid (±8×), with black diamonds marking the (off-center) sampling center, red × the true D^*, and blue + the inferred D^*. Top right: extrapolation error in compute-optimal token prediction at 10^{24} FLOPs for each grid width. Bottom row: exponent and intercept errors across grid widths from XS (±2×) to XL (±16×), plotted on the same y-axis scale. The exponent is recovered perfectly (flat at zero) while the intercept shows systematic bias that varies with grid width.

into the fit, partially compensating for the misplaced center.

4.3.2 Drifting Bias

When the offset varies with compute budget, a different failure mode occurs, and we illustrate this by applying a multiplicative drift. The sampling center starts at the true optimum for the lowest budget and drifts to 3× the true optimum at the highest budget, interpolating linearly in log-compute space.

Because the offset now differs across compute budgets, it no longer cancels in the slope of $\log D^*$ vs $\log C$. Both the exponent and the intercept are affected, as seen in Figure 7. The bottom-left panels of Figures 6 and 7 highlight the most important difference between compute-independent (constant bias) and compute-dependent offsets (drift bias).

4.4 Real IsoFLOP Curves

The previous sections used synthetic, noise-free simulations to isolate Approach 2's biases under controlled conditions. A good question is whether the conditions that trigger these biases, asymmetric loss surfaces and imperfectly centered sampling, actually arise in practice. To get a sense of

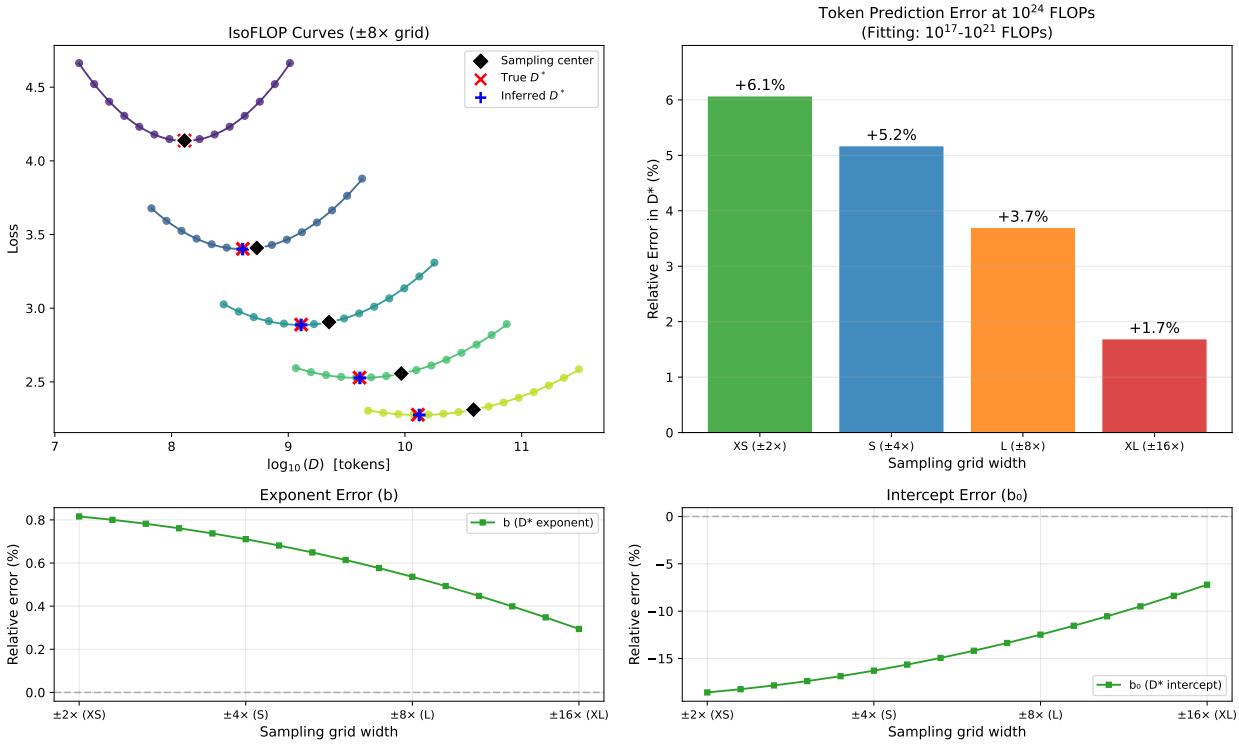


Figure 7: Effect of a linear drift in sampling centers (centered at true optimum for lowest budget, drifting to 3× at highest budget) on the symmetric surface. Unlike the constant bias case, the exponent error (bottom left) is now non-zero: the slope of $\log D^*$ vs $\log C$ is distorted because the offset varies across compute budgets.

this, we can look at IsoFLOP curves published in three[4] prominent scaling law studies [Hoffmann et al., 2022, Grattafiori et al., 2024, DeepSeek-AI et al., 2024] in Figure 8.

[4]These three and 10+ more can be found at https://docs.google.com/document/d/1Ab3cYJJYu7t_99Idj9gpHv2icZmRTPBJ_brJxsgnVA4.

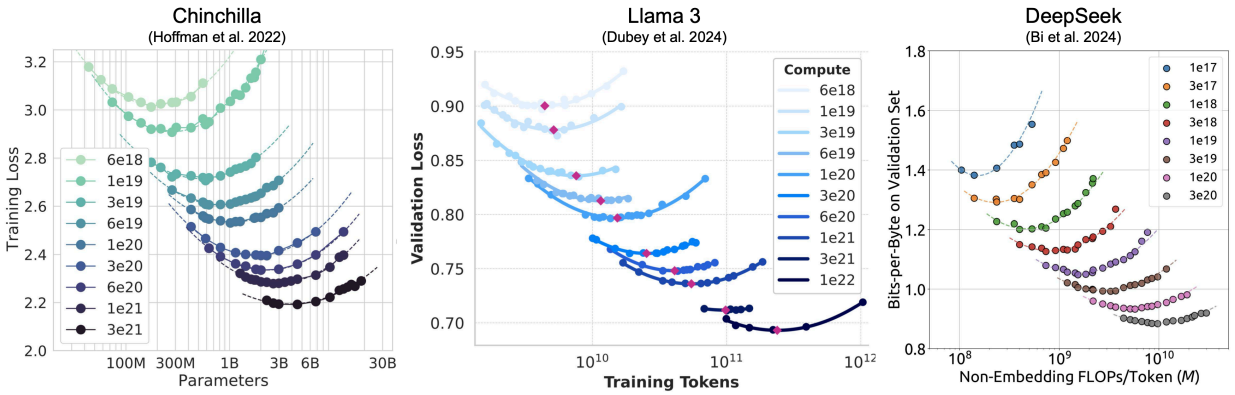


Figure 8: IsoFLOP curves from three published scaling law studies. Left: Chinchilla (loss vs parameters). Center: Llama 3 (loss vs tokens). Right: DeepSeek (bits-per-byte vs parameters). Each panel shows curves at multiple compute budgets, fit using Approach 2.

We first consider biases that might arise from asymmetry. Published scaling law studies frequently report unequal scaling exponents, as shown in Table A2. Several reported allocation-exponent point estimates reach the levels of imbalance used in our simulations. These empirical allocation ratios need not identify loss-surface exponent ratios when compute conventions or fitting procedures differ, and uncertainty can include equal exponents.

Beyond surface asymmetry, the IsoFLOP curves in Figure 8 also show visible signs of off-center sampling and drift:

- **Off-center sampling:** At some compute budgets, the sampling grid does not appear centered at the curve minimum, placing more points on one side of the optimum than the other.
- **Drifting centers:** The degree of off-centering appears to vary across compute budgets rather than remaining constant, which is the drifting-bias pattern that distorts both exponents and intercepts.

To be clear, this is not a criticism of these studies. These are among the most careful and influential scaling law analyses yet published. The point is a more general one: the conditions under which Approach 2's biases take effect, asymmetric surfaces and imperfect sampling centers, can arise in practice. These selected examples do not establish their prevalence or quantify the bias in the original studies.

4.4.1 Compounding Errors

Given evidence that both surface asymmetry and off-center sampling are present in real studies, we can simulate what happens when these biases act simultaneously. Using the same three loss surfaces from earlier sections, we combine them with the 3× drift and 3× constant offset from the off-center analysis. We fit Approach 2 on compute budgets from 10^{17} to 10^{21} FLOPs and extrapolate D^* predictions to 10^{24} FLOPs across all four grid widths.

Results from this simulation are shown in Figure 9. Compared to the baseline in Figure 5, where asymmetry bias alone produces errors up to −5% on Chinchilla and −23% on the Asymmetric surface, the two bias sources interact in opposite directions. Off-center sampling pushes errors positive (overestimating D^*), while asymmetry bias pushes errors negative (underestimating). When the two sources reinforce rather than cancel, errors grow substantially—reaching nearly +35% at a maximum on the Asymmetric surface.

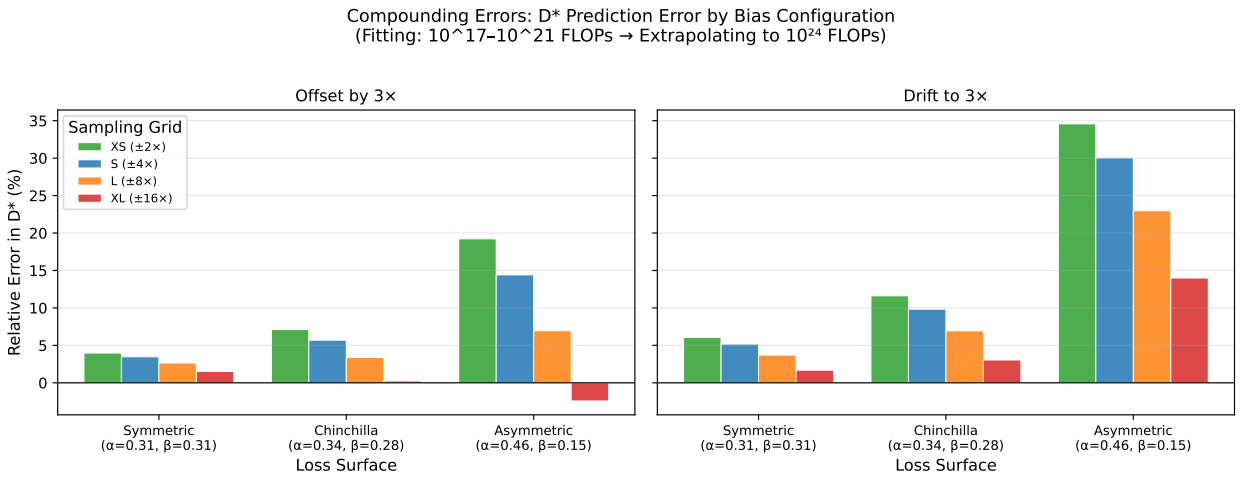


Figure 9: Relative error in D^* at 10^{24} FLOPs with off-center sampling on all three loss surfaces. Left: constant 3× center offset at every budget. Right: linear drift to 3× at the highest compute budget. Bars are grouped by sampling grid width (XS through XL). Negative values indicate underestimation; positive values indicate overestimation. On the symmetric surface, the constant offset results correspond to Figure 6 and the drift results correspond to Figure 7; the asymmetric surfaces reveal how these sampling biases interact with the inherent asymmetry bias.

Figure A4 provides a more detailed view of these results. It shows how D^* extrapolation errors change across compute budgets, which sources of bias grow with extrapolation distance (drift), which remain constant (surface asymmetry and constant offsets), and how these patterns vary across two drift rates and center offset magnitudes.

In summary, multiple sources of bias act simultaneously in any real experiment. Surface asymmetry and off-center sampling can each produce meaningful errors on their own. When they happen to act in the same direction, the combined error can compound. On our hypothetical Asymmetric surface with drift to 3×, errors reach 35% even when using the narrowest grid where the parabolic approximation is most accurate. When they oppose, partial cancellation can occur, but this depends on the specific combination of surface geometry, offset magnitude, and grid width.

5 Robust Fits: Unbiased Estimation with Linear Separation

The previous sections showed that Approach 2's parabolic approximation introduces systematic biases in intercepts (from asymmetry) and potentially exponents (from off-center sampling), and that the conditions driving these biases are visible in published scaling law studies. The natural alternative is Approach 3, which fits all five surface parameters (E, A, B, α, β) simultaneously via nonlinear optimization. This avoids the parabolic approximation but introduces other problems.

5.1 Problems with Direct Surface Fitting

A survey of 51 scaling law papers [Li et al., 2025b] documents the landscape of fitting practices and their failure modes. The problems described within it apply to scaling laws in general, not just Chinchilla forms like Approach 3. The survey reports 23 papers without a description of the optimization process and documents missing optimizer, loss-function, and initialization details. These omissions compound reproducibility challenges, while reported procedures suggest several common challenges.

First, the most frequently used optimizers for scaling law fits are BFGS and L-BFGS. Some studies use SGD-family optimizers like Adam and Adagrad, though these are noted as sometimes poorly suited for curve fitting due to limited data efficiency. BFGS builds an inverse-Hessian approximation from changes in gradients; it does not require evaluating second derivatives. L-BFGS stores a limited history of updates to reduce memory use.[5] The survey also identifies a study that uses pure grid search to fit its scaling-law parameters [Goyal et al., 2024].

Second, the survey reports sensitivity to initialization in nonlinear fits (Sections 6 and 7.4). Mitigations for this include grid search over hundreds or thousands of starting points (running the optimizer from each and keeping the best fit), random sampling of starting points, evaluating a coarse grid without optimization and seeding the optimizer from the single best candidate, or initializing from previously published parameter values.

Third, numerical stability is a significant issue that can be framed more concretely within the context of some of the simulations discussed in prior sections. For example, consider the Hessian of the residual sum of squares (RSS) objective for a five-parameter fit on noise-free data from the hypothetical Asymmetric surface ($\alpha = 0.465$, $\beta = 0.155$), using five IsoFLOP contours from 10^{17} to 10^{21} FLOPs with 15 points per curve. Its eigenvalues quantify how sensitive the objective is to perturbations along each parameter direction [Ghorbani et al., 2019], and the condition number κ (the ratio of the largest to the smallest eigenvalue) measures how difficult the landscape is for gradient-based methods to navigate. For this surface, the five eigenvalues span from approximately 8×10^{-6} to 3×10^{6}, giving $\kappa \approx 3.5 \times 10^{11}$. The two flattest directions (smallest eigenvalues) point almost entirely along the linear coefficients A and B. Near the optimum, perturbing either coefficient barely changes the RSS, making them effectively underdetermined by the data even when the data are perfect. The steepest directions are dominated by the scaling exponents α and β, and this difference in scale between exponents and linear coefficients is likely the primary source of numerical instability for most optimizers.

Quasi-Newton methods like L-BFGS build an approximate inverse Hessian to scale gradient steps across parameter directions. When eigenvalues span 12 orders of magnitude, the gradient signal along the flat A/B directions is negligible compared to the steep α/β directions, and convergence criteria are often satisfied by progress in the steep directions before the flat directions are resolved. Separating the linear parameters from the nonlinear search eliminates the ill-conditioned directions entirely. The resulting two-dimensional landscape over (α, β) has a Hessian condition number of $\kappa \approx 11$ in our example, a reduction by a factor of roughly 3×10^{10}. This motivates an algorithm that exploits the partially linear structure of the Chinchilla loss surface to search only the well-conditioned two-dimensional subspace.

5.2 Variable Projection (VPNLS)

The Chinchilla loss surface has a partially linear structure that can be exploited using the established variable projection method [Golub and Pereyra, 1973]. For any fixed values of α and β, the remaining parameters (E, A, B) enter the model linearly and can be solved exactly via least squares. This is the same computational shortcut that motivates Approach 2 (optimizing exponential terms separately from linear terms), but applied here without the parabolic approximation.

The algorithm searches over (α, β) and, at each candidate pair, solves for (E, A, B) via least squares. A coarse $k \times k$ grid search identifies a good starting region, and a local optimizer refines it. The linear separation is maintained throughout. The optimizer only ever navigates the two-dimensional (α, β) surface, never the full five-parameter space. We term this method Variable Projection with Non-negative Least Squares (VPNLS).

[5] https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html

In its simplest form, VPNLS uses non-negative least squares (NNLS) [Lawson and Hanson, 1995] for the inner solve and a gradient-free optimizer like Nelder-Mead for the outer search:

function VPNLS(data):

  function objective(α, β):
    $\mathbf{X} \leftarrow [\mathbf{1},\ N^{-\alpha},\ D^{-\beta}]$ // *design matrix*
    $(E, A, B) \leftarrow \text{NNLS}(\mathbf{X}, L)$ // *linear solve,* $E, A, B \geq 0$
    return $\|L - \mathbf{X} \cdot [E, A, B]\|^2$

  $(\alpha_0, \beta_0) \leftarrow \arg\min \text{objective}(\alpha,\ \beta)$ // *coarse* $k \times k$ *grid*
  $(\alpha^*, \beta^*) \leftarrow \text{NelderMead}(\text{objective},\ \text{start}{=}(\alpha_0, \beta_0))$ // *refine in 2D*
  $(E^*, A^*, B^*) \leftarrow \text{NNLS}(\mathbf{X}(\alpha^*, \beta^*), L)$ // *recover linear params*

  return $(E^*, A^*, B^*, \alpha^*, \beta^*)$

Switching the inner solve from NNLS to ordinary least squares (OLS) makes the objective differentiable and enables analytical gradients. By the envelope theorem, the gradient of RSS with respect to (α, β) depends only on the current residuals and design matrix, not on implicit derivatives of the optimal (E, A, B). This allows L-BFGS-B to be used with exact gradients rather than finite differences:

function VPNLS(data):

  function objective_and_grad(α, β):
    $\mathbf{X} \leftarrow [\mathbf{1},\ N^{-\alpha},\ D^{-\beta}]$ // *design matrix*
    $(E, A, B) \leftarrow \text{OLS}(\mathbf{X}, L)$ // *linear solve*
    $r \leftarrow L - \mathbf{X} \cdot [E, A, B]$ // *residual vector*
    $\partial\text{RSS}/\partial\alpha \leftarrow 2A \cdot r^\top(\log(N) \odot N^{-\alpha})$
    $\partial\text{RSS}/\partial\beta \leftarrow 2B \cdot r^\top(\log(D) \odot D^{-\beta})$
    return $\|r\|^2$, $[\partial\text{RSS}/\partial\alpha,\ \partial\text{RSS}/\partial\beta]$

  $(\alpha_0, \beta_0) \leftarrow \arg\min \text{objective}(\alpha,\ \beta)$ // *coarse* $k \times k$ *grid*
  $(\alpha^*, \beta^*) \leftarrow \text{L-BFGS-B}(\text{objective_and_grad},\ \text{start}{=}(\alpha_0, \beta_0))$ // $\alpha, \beta > 0$
  $(E^*, A^*, B^*) \leftarrow \text{OLS}(\mathbf{X}(\alpha^*, \beta^*), L)$ // *recover linear params*

  return $(E^*, A^*, B^*, \alpha^*, \beta^*)$

Because the nonlinear search is over exponential terms only, grid density scales with the number of exponential terms rather than the total number of parameters. A 32^2 grid over (α, β) provides 1,024 candidates with fine resolution, whereas the same budget in 5D gives only $4^5 = 1{,}024$ points spread thinly across all parameters.

The implementation of this algorithm used in the remainder of this work was validated against `ml-scalefit` [Shukor et al., 2025a]. See Appendix A for details.

5.3 Method Comparison (Parameter Recovery)

We compare six method configurations on noise-free synthetic data across three loss surfaces (symmetric, Chinchilla, and high imbalance) and 20 sampling ranges in Figure 10.

The configurations fall into two groups. The first uses 5D direct optimization (Approach 3), fitting all five parameters jointly with L-BFGS-B using either analytical or numerical (3-point central difference) gradients. The second uses 2D variable projection over (α, β) only, comparing L-BFGS-B with analytical gradients, L-BFGS-B with numerical gradients, Nelder-Mead (gradient-free, NNLS inner solve), and a fine 256^2 grid search with no local refinement. Both groups use

the same total initialization budget: the 5D methods search a $4^5 = 1{,}024$-point grid over all five parameters, while the 2D methods search a $32^2 = 1{,}024$-point grid over (α, β) only, so that accuracy differences reflect the optimizer and loss-landscape geometry rather than an initialization advantage.

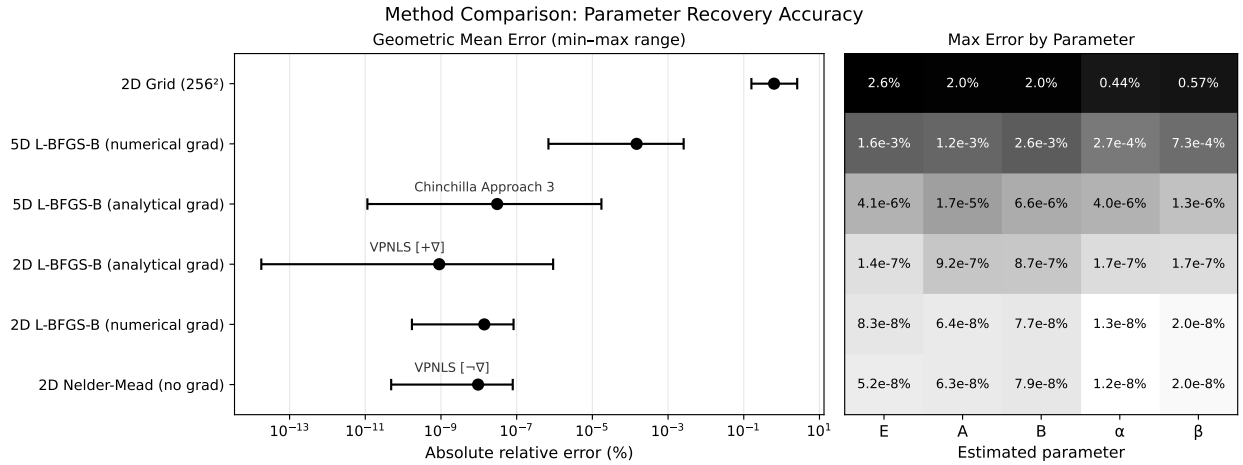


Figure 10: Comparison of six fitting methods on noise-free synthetic data across three loss surfaces and 20 sampling ranges (60 fits total per method). Left: geometric mean of |relative error| (%) pooled across all surfaces, grid widths, and parameters, with horizontal bars spanning the min-to-max range. Right: maximum |relative error| (%) per parameter, on a log-scale colormap. Methods are sorted by geometric mean error, with the worst at top. VPNLS [+∇] uses analytical gradients (L-BFGS-B with OLS inner solve); VPNLS [¬∇] is gradient-free (Nelder-Mead with NNLS inner solve).

All three locally optimized 2D variable projection methods (L-BFGS-B with analytical gradients, L-BFGS-B with numerical gradients, and Nelder-Mead) recover parameters to machine precision ($\sim 10^{-7}$% or better) across all surfaces. The well-conditioned 2D landscape makes even finite-difference gradients reliable in the reduced search space. High-resolution grid search (256^2) is stable but limited by grid resolution, providing the poorest precision among the 2D methods.

5D direct optimization (Approach 3) with analytical gradients also achieves high precision in this noiseless setting (errors on the order of 10^{-5}%), though with larger errors than any 2D method, particularly on the Asymmetric surface. The 5D numerical gradient variant performs worse, with errors on the order of 10^{-3}%. Figure A3 breaks this down by surface and sampling range.

5.4 Method Comparison (Exponent Inference)

The parameter recovery results above are noise-free, which is obviously not representative of practice. We now extend the compounding errors scenario (Section 4.4.1; Asymmetric surface with a 3× drift at the ±8× grid) to a statistical setting. Gaussian noise is added to loss values at three levels ($\sigma = 0.05, 0.1, 0.2$), applied uniformly across all compute budgets. Figures A8 and A9 provide empirical support for this uniform treatment, finding no strong evidence that residual variance differs by budget in most real IsoFLOP experiments. The number of compute budgets varies from 2 to 4, and the number of points per IsoFLOP curve ranges from 4 to 32. Each configuration is repeated with 256 independent noise realizations, yielding 9,216 fits per method and 46,080 fits in total. Figure A2 shows what the noisy IsoFLOP samples look like at each noise level.

Because a scaling law study is typically run once rather than repeated hundreds of times, sporadic optimizer failures that produce large errors in a minority of fits are arguably the most

consequential practical risk. The comparison below emphasizes maximum errors alongside typical accuracy for this reason.

Five methods are compared in Figure 11. Approach 2 uses the parabolic IsoFLOP pipeline. Naive Approach 3 uses a single random initialization without LSE reparameterization. MLE Approach 3 uses grid initialization with the direct MSE objective (the maximum likelihood estimator under additive Gaussian noise). The final Approach 3 variant uses grid initialization with the LSE reparameterization and log-loss objective, matching the formulation in the Chinchilla paper [Hoffmann et al., 2022]. VPNLS uses L-BFGS-B with analytical gradients.

The parameter recovery comparison (Section 5.3) evaluated all five surface parameters, but Approach 2 does not estimate them individually. Here we focus on the scaling exponents $a = \beta/(\alpha+\beta)$ and $b = \alpha/(\alpha+\beta)$, which determine compute-optimal allocation and are the quantities all five methods can be compared on directly. Figure 11 pools these exponent errors across all experimental conditions.

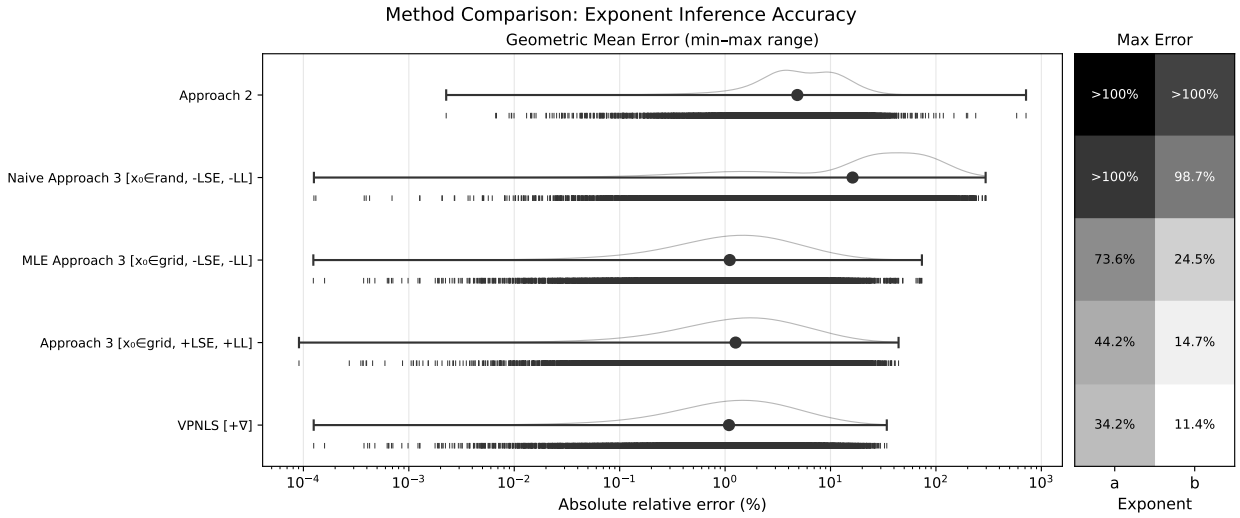


Figure 11: Comparison of five fitting methods on noisy synthetic data, pooled across three noise levels, three budget counts, four points-per-curve settings, and 256 noise realizations (9,216 fits per method). Left: geometric mean of |relative error| (%) with min-to-max range bars, rug ticks for individual errors, and kernel density estimate. Right: maximum |relative error| (%) per exponent. Methods sorted by worst-case error, worst at top. Surface: Asymmetric ($\alpha = 0.465$, $\beta = 0.155$) with 3× drift, ±8× grid, compute budgets 10^{17}–10^{21} FLOPs.

Approach 2 has the largest maximum errors overall (716% on a, 239% on b), reflecting the structural bias from the parabolic approximation (Section 4.2). Even with 32 points per curve and low noise, the systematic inaccuracy persists. Naive Approach 3 (single random initialization, no LSE) also produces extreme maximum errors (296% on a), confirming that without careful initialization and parameterization, 5D optimization is unreliable.

The canonical Chinchilla Approach 3 formulation (LSE reparameterization with log-loss objective, grid initialization) reduces maximum errors dramatically to 44% on a. The LSE reparameterization enforces positivity and stabilizes optimization. However, because the noise model in these simulations is additive Gaussian, the log-loss objective is slightly misspecified, which is visible as a higher geometric mean error compared to the MLE variant and VPNLS.

The MLE variant of Approach 3 (grid initialization, original-space MSE) is the correct maximum likelihood estimator for additive Gaussian noise. In this experiment, however, it exhibited

larger maximum errors than either canonical Approach 3 or VPNLS (74% vs 44% vs 34% on the a exponent), despite slightly better typical accuracy. This is primarily due to the lack of LSE reparameterization.

VPNLS produced the smallest maximum errors in this experiment (34% on a, 11% on b) with typical accuracy roughly equivalent to Approach 3. The detailed breakdown in Figure A5 shows how errors distribute across noise levels, budget counts, and dataset sizes.

5.5 Method Comparison (Data Efficiency)

The exponent inference comparison above used an asymmetric surface with off-center sampling, conditions designed to test structural biases. Here we do the opposite and build on the ideal conditions of the symmetric surfaces section ($\alpha = \beta = 0.31$, perfectly centered sampling) where Approach 2 has no structural bias. All methods are unbiased in this setting, so the relevant comparison is variance. For unbiased estimators, statistical efficiency is proportional to the inverse of variance, so lower variance means more precise estimates for a given sample size. Noise levels ($\sigma \in \{0.01, 0.02, 0.05\}$) are calibrated to the empirical residual standard deviations observed across real IsoFLOP experiments (Figures A8 and A9). Even under these ideal conditions, Figure 12 shows that Approach 2 recovers scaling exponents with roughly 8× higher variance than Approach 3 or VPNLS.

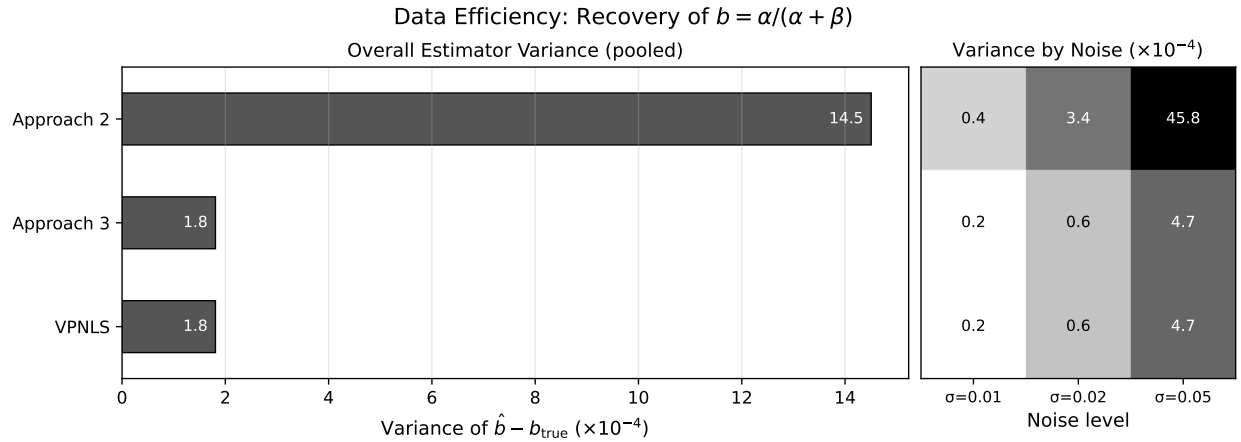


Figure 12: Estimator variance for the compute-optimal exponent $b = \alpha/(\alpha + \beta)$ on a symmetric surface ($\alpha = \beta = 0.31$, $A = B = 400$, $E = 1.69$, true $b = 0.5$) with perfectly centered IsoFLOP sampling. Left: overall variance pooled across noise levels. Right: variance by noise level. Sweep over points per curve $\in \{21, 31, 41\}$, budget counts $\in \{3, 5, 7\}$, log-widths $\in \{\pm 2\times, \pm 4\times, \pm 8\times\}$, noise levels $\sigma \in \{0.01, 0.02, 0.05\}$, with 10 trials per configuration. Lower variance is better. Approach 3 and VPNLS achieve comparable precision; Approach 2 is consistently less efficient.

Figure A6 breaks these results down by noise level, showing the full distribution of signed errors for each method.

6 Conclusion

The Approach 2 biases documented in this paper are structural, not statistical. They exist on noise-free data with perfect experimental conditions and, as the noisy method comparison (Section 5.4) shows, persist under realistic noise levels with varying amounts of data.

At least three sources of Approach 2 error can compound in practice. IsoFLOP sampling grid width, uncentered IsoFLOP sampling, and loss surface asymmetry all bias inference and extrapolations in different ways. Published IsoFLOP curves from prominent scaling law studies (Section 4.4) show clear signs of both asymmetry and off-center sampling. At frontier compute scales (Section 3), we estimate that these biases translate to a potential 6.5% decrease in training FLOPs ($1.4M) on Llama 3 data and potentially more on multimodal surfaces with greater asymmetry.

Approach 3 eliminates these biases. With grid initialization and LSE reparameterization (as specified in the original Chinchilla paper [Hoffmann et al., 2022]), it achieves typical accuracy comparable to VPNLS in our experiments. A recent survey [Li et al., 2025b] suggests that these important optimization details may be omitted or not reported in some studies.

VPNLS is at least as stable and accurate as more typical Approach 3 implementations while being structurally simpler. Variable projection separates exponential from linear terms, reducing the nonlinear search to the exponential terms only. This makes dense grid search practical because the grid grows with the number of exponential terms rather than all parameters, and these exponents occupy tight ranges (typically 0 to 1) unlike the linear coefficients which span several orders of magnitude.

This structural advantage extends naturally to richer loss surface specifications. The analytical extensions discussed in the introduction (Section 1), such as epochs, data quality, and MoE sparsity, often add linear terms that could be omitted from direct optimization. This structural simplification may provide a more scalable foundation for future work. It also lends well to simple implementations of the Chinchilla loss model alone, e.g. it is possible to express a simplified version of the algorithm[6] that exhaustively evaluates a bounded grid over $\alpha, \beta \in (0, 1]$ with an OLS inner solve and no optimization or linear-algebra dependencies. This selects the lowest evaluated grid objective; it does not certify a global optimum over the continuous parameter space.

Lastly, practitioners using Approach 2 should be aware that scaling exponent estimates carry a systematic bias that often grows with loss surface asymmetry, sampling center offsets, and sampling grid width. When fitting surfaces directly with Approach 3 instead, grid initialization and LSE reparameterization should be used. And VPNLS offers equivalent accuracy with a simpler optimization structure to aid in scaling this approach to richer formulations, or as a simpler alternative to Approach 2.

6.1 Limitations

Several known limitations qualify the conclusions of this study:

- **Irreducible loss dominance at large scale.** At sufficiently large compute budgets, scaling properties are dominated entirely by the irreducible loss E. When token counts and model sizes at fixed compute budgets are large enough, the Chinchilla surface reaches E asymptotically and all training configurations become equally effective, meaning that extrapolations are irrelevant and compute-optimal training is no longer informed by scaling laws. We assume this study is only relevant to practitioners working in a regime where downstream model quality can still effectively be informed by scaling law extrapolations per the Chinchilla model.

- **Assumed correctness of the Chinchilla loss surface.** We assume the Chinchilla loss surface model $L(N, D) = E + A/N^{\alpha} + B/D^{\beta}$ is correct in practice. Empirical work supports the robustness of Chinchilla allocation prescriptions [Schaeffer et al., 2025], but this does not establish the correctness of the functional form in every setting. Alternatives include the

[6] https://gist.github.com/eric-czech/ebd9a80d58c7b5e9c40ba390ff884617

Kaplan loss model [Kaplan et al., 2020], refined analytical surfaces like Farseer [Li et al., 2025a] and MuPT [Qu et al., 2024], and agent-discovered functional forms [Lin et al., 2025].

- **Qualitative characterization of published study errors.** Likely errors in published studies are characterized qualitatively rather than quantified. We believe the qualitative characterization is compelling enough on its own to justify that real IsoFLOP sampling pathologies occur in practice, but they are difficult to quantify precisely because they do not follow the convenient theoretical model we use for those pathologies in our simulations.

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Appendix

A VPNLS Implementation Validation

We validated our VPNLS implementation against `ml-scalefit` [Shukor et al., 2025a]. Both were run on the Chinchilla dataset (217 training points with $C < 10^{21}$, inputs normalized by $N/10^6$ and $D/10^9$). Table A1 shows the fitted parameters.

Both Approach 3 variants use log-sum-exp (LSE) reparameterization to ensure non-negativity of the additive loss components. They differ only in whether the objective is evaluated on raw loss values (linear) or their logarithm (log-scaled), as specified in the original Chinchilla formulation.

The `ml-scalefit` MSE configuration matches VPNLS to three decimal places (1×10^{-3}). Approach 3 with LSE reparameterization but without log-scaled loss also produces identical results. Only when log-scaled loss is used do the estimates diverge. This highlights that both VPNLS and `ml-scalefit` (with MSE) are maximum likelihood estimators under additive Gaussian error on the raw loss, not a multiplicative error model as implied by the original Chinchilla Approach 3 log-loss formulation. The Huber variant down-weights outliers and produces modestly different estimates.

Method	E	A	B	α	β	a	b
VPNLS	1.9051	4.0005	1.0509	0.3511	0.4587	0.5665	0.4335
Approach 3 (log-scaled loss)	1.8926	4.0146	1.0571	0.3522	0.4430	0.5571	0.4429
Approach 3 (linear loss)	1.9051	4.0005	1.0509	0.3511	0.4587	0.5665	0.4335
`ml-scalefit` (MSE)	1.9051	4.0001	1.0509	0.3510	0.4588	0.5665	0.4335
`ml-scalefit` (Huber)	1.8267	3.7066	1.0386	0.3240	0.4080	0.5573	0.4427

Table A1: VPNLS validation against `ml-scalefit` on the Chinchilla dataset. VPNLS and `ml-scalefit` (MSE) agree to three decimal places (1×10^{-3}). Approach 3 with linear loss is identical to VPNLS; only the log-scaled loss variant diverges.

B IsoFLOP Quality Control Pipeline

The IsoFLOP quality control pipeline applies these steps in this order:

- **Pre-QC** removes duplicate and near-duplicate parameter values within each budget. Exact duplicates are disambiguated by keeping the point whose implied compute ($6ND$) is closest to the nominal budget (ties broken by minimum loss). Near-duplicates are identified by greedy binning in log-parameter space (tolerance 0.01) and disambiguated the same way. Budgets with fewer than 6 remaining points are then removed entirely.
- **Off Center** removes points that fall outside a symmetric window around each IsoFLOP curve's parabola minimum (2.5× the distance from the minimum to the nearer edge), targeting the off-center sampling bias identified in the main text. An Akima spline step also removes individual statistical outliers at this stage by fitting a leave-one-out spline to non-outlying points at each budget and flagging those with extreme MAD-based residuals. Across all experiments, every spline-identified point (28 of 28) is terminal, having no non-outlying point on at least one side in sorted parameter order. In practice, the spline step acts as a more aggressive complement to the conservative off-center filter.
- **Weak Curvature** first flags entire budgets with negative parabola curvature (quadratic coefficient ≤ 0). For remaining budgets, it checks whether the curvature is significantly positive by testing whether the 95% confidence interval on the quadratic coefficient includes zero. Budgets that fail either check are removed.
- **Post-QC** removes budgets that fell below the 6-point minimum after earlier filtering.

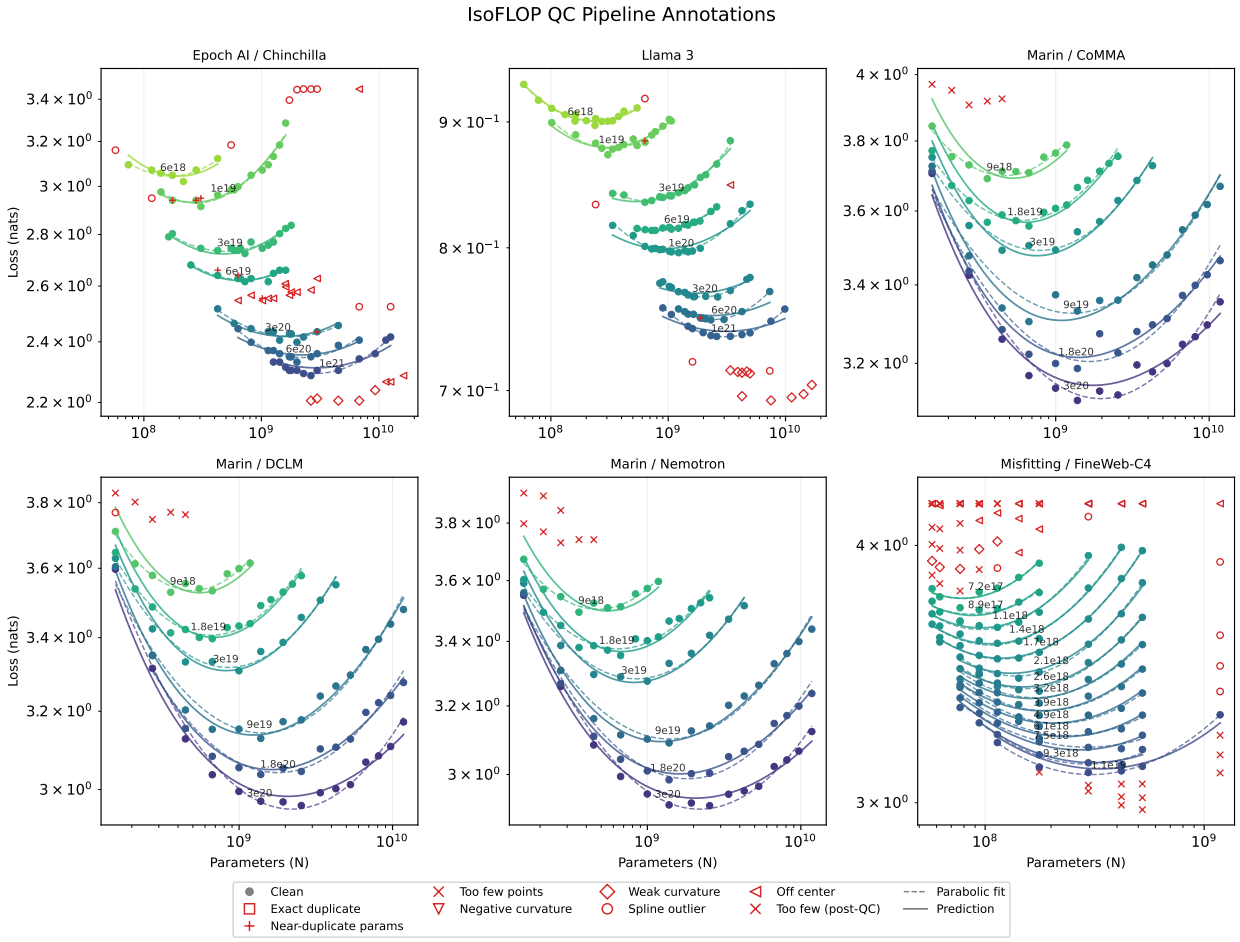


Figure A1: IsoFLOP curves for six experiments spanning different datasets and model families. Clean points (circles) and outlier points (markers by reason) are shown alongside parabolic fits (dashed) and Approach 3 predictions with per-budget FLOP factor adjustment (solid). Experiments shown: Epoch AI/Chinchilla, Llama 3, Marin/CoMMA, Marin/DCLM, Marin/Nemotron, and Misfitting/FineWeb-C4.

C IsoFLOP Samples with Noise

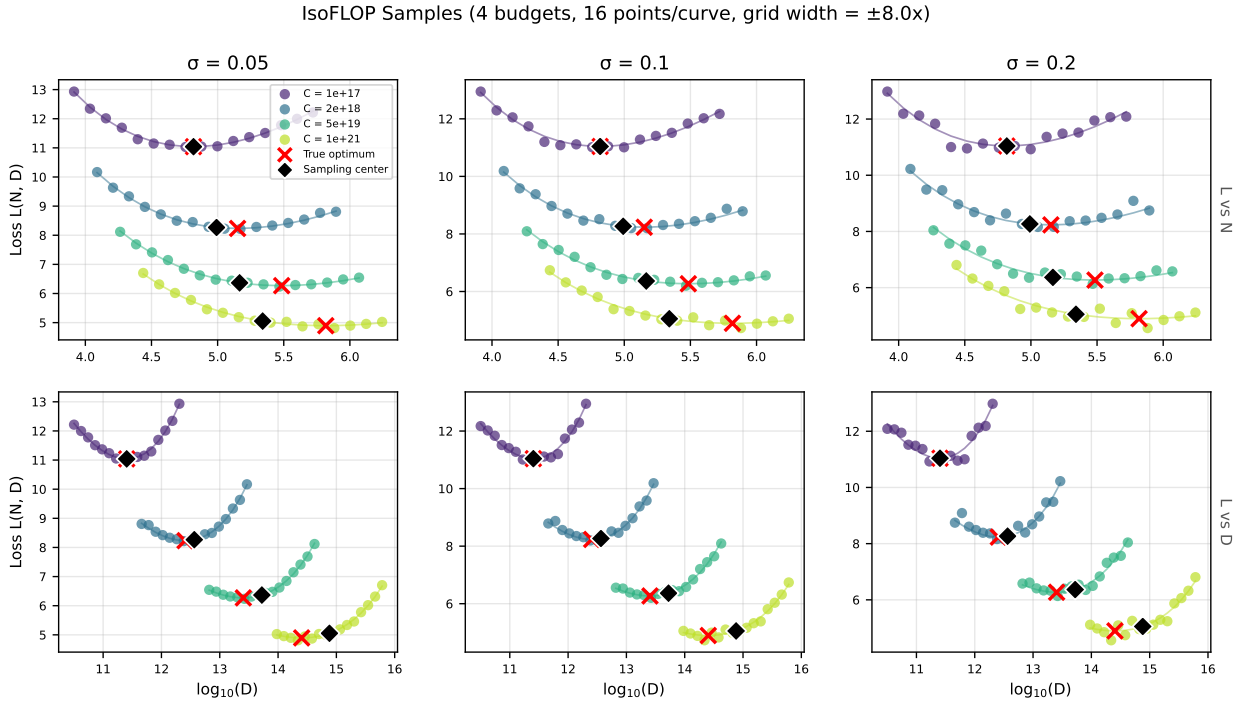


Figure A2: Noisy IsoFLOP samples used in the exponent inference comparison (Figure 11). Columns correspond to noise levels ($\sigma = 0.05, 0.1, 0.2$); rows show loss versus $\log_{10}(N)$ (top) and $\log_{10}(D)$ (bottom). Scatter points are noisy observations; solid curves show the noiseless reference surface. Red $\times$ marks the true compute-optimal point at each budget; black diamonds mark the sampling centers, which drift away from the true optima at higher compute budgets. Surface: Asymmetric ($\alpha = 0.465$, $\beta = 0.155$), 4 budgets (10^{17}–10^{21} FLOPs), 32 points per curve, $\pm 8\times$ grid.

D Detailed Method Comparison

Full per-parameter, per-surface, per-sampling-range error breakdown for all six method configurations.



Figure A3: Per-parameter recovery error for six fitting methods across three loss surfaces and 20 sampling ranges (baseline, no bias). Each panel shows absolute relative error (%) on a log scale versus sampling range, with one curve per method. Rows correspond to loss surfaces (symmetric, Chinchilla, high imbalance); columns correspond to parameters (E, A, B, α, β).

E Combined Extrapolation Error by Compute Budget

Detailed view of D^* extrapolation error as a function of compute budget, showing how errors evolve from 10^{22} to 10^{25} FLOPs across sampling ranges, loss surfaces, and bias configurations.

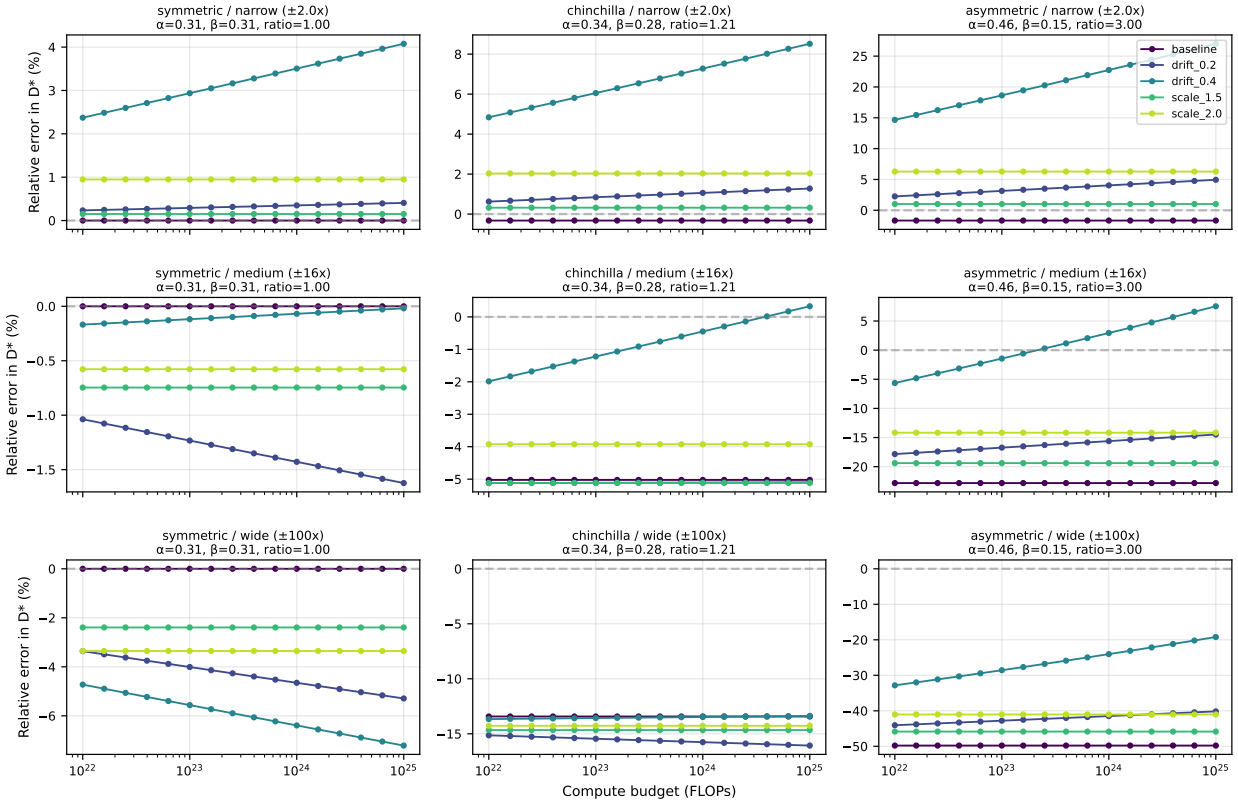


Figure A4: Relative error in compute-optimal token count D^* when extrapolating from the fitting range (10^{17}–10^{21} FLOPs) to higher compute budgets (10^{22}–10^{25} FLOPs), with asymmetry and sampling biases acting simultaneously. Columns correspond to loss surfaces (symmetric, Chinchilla, Asymmetric); rows correspond to sampling ranges (narrow ±2×, medium ±16×, wide ±100×). The wide row uses an extreme grid width not employed in the main text, included here to further illustrate how far results can deviate with a misconfigured experiment. Each curve represents a different sampling bias configuration: baseline (no bias), two linear drift rates (drift_0.2 and drift_0.4, where the value is the $\log_{10}$ offset at the highest compute budget), and two constant center offsets (scale_1.5 and scale_2.0, where the value is the multiplicative factor applied at every budget). On symmetric surfaces, errors are driven entirely by off-center sampling; on asymmetric surfaces, the inherent surface bias adds a constant offset visible as the non-zero baseline curve. Drift-based biases produce errors that grow with extrapolation distance (steeper curves), while constant offsets and surface asymmetry produce flat errors.

F Exponent Inference Error Breakdown

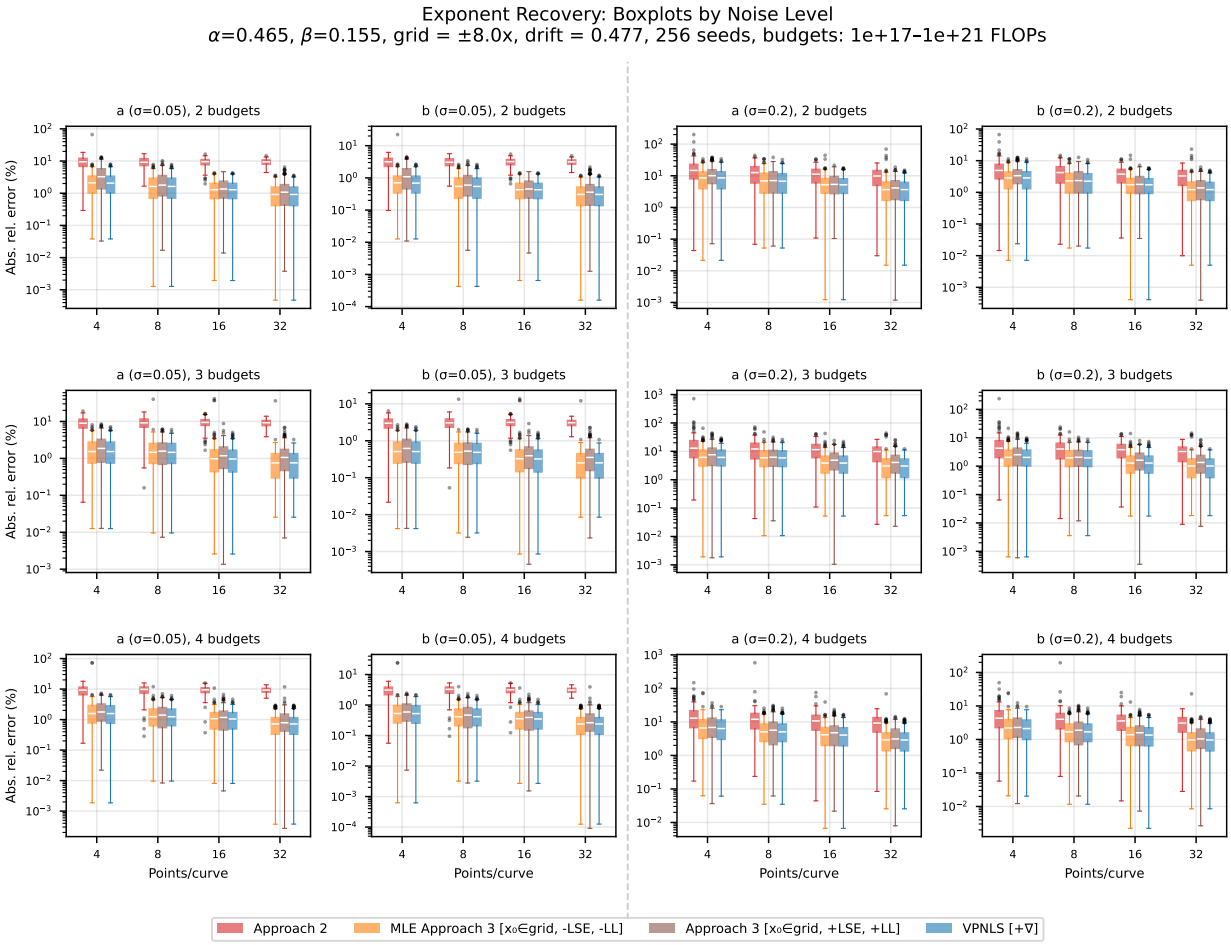


Figure A5: Per-condition breakdown of exponent inference errors from Figure 11, showing 4 of 5 methods (excludes Naive Approach 3, which serves only as a negative control in the main figure). Rows correspond to number of compute budgets (2, 3, 4); left columns show the lowest noise level ($\sigma = 0.05$) and right columns show the highest ($\sigma = 0.2$), each split by exponent (a and b). Within each panel, boxplots show absolute relative error (%) on a log scale for each method at each points-per-curve setting (4, 8, 16, 32), over 256 noise realizations.

G Data Efficiency Error Breakdown

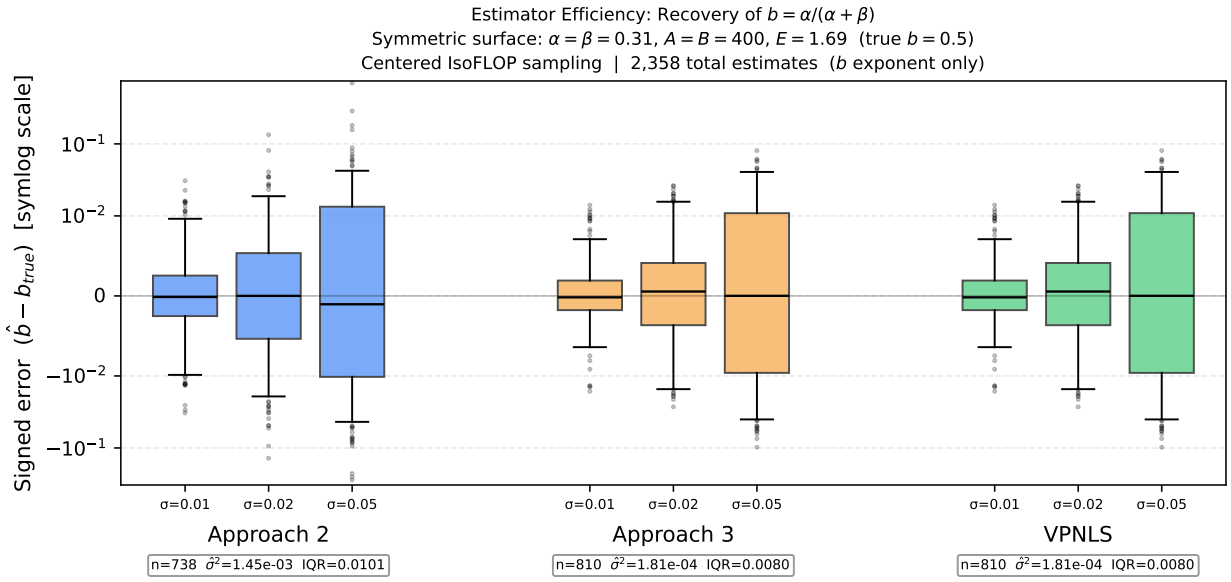


Figure A6: Per-noise-level breakdown of signed exponent errors from Figure 12. Boxplots show the distribution of $\hat{b} - b_{\text{true}}$ on a symlog scale for each method at each noise level. Pooled summary statistics (sample size, variance, IQR) are shown below each method group.

H Published Scaling Exponents

Table A2 collects published compute-allocation exponents a and b (where $N^* \propto C^a$ and $D^* \propto C^b$) from scaling law studies across a range of domains, architectures, and fitting methods.

Source	Domain	Config	Method	a	b	Ratio
DeepSeek [DeepSeek-AI et al., 2024]	Web text	OpenWebText2‡	Appr. 2	0.73	0.27	2.70
		MassiveText‡		0.49	0.51	0.96
		DeepSeek (early)		0.450	0.550	0.82
		DeepSeek (current)		0.524	0.476	1.10
		DeepSeek (OWT2)		0.578	0.422	1.37
EHR [Zhang et al., 2025]	Health records	Transformer	Appr. 2	0.58	0.44	1.32
Evo [Nguyen et al., 2024]†	DNA sequences	Transformer++	Appr. 2	0.552	0.551	1.00
		Mamba		0.388	0.487	0.80
		Hyena		0.504	0.499	1.01
		StripedHyena		0.483	0.539	0.90
IL [Tuyls et al., 2024]	Games (IL)	NetHack	Appr. 2	0.61	0.39	1.56
		Battle Zone		0.58	0.51	1.14
		Breakout		0.74	0.46	1.61
DiT [Liang et al., 2024]	Image-text	Diffusion Trans.	Appr. 2	0.568	0.432	1.32
DLM [von Rütte et al., 2025]	Web text	Uniform diffusion	Appr. 2	0.589	0.411	1.43
Audio [Manakul et al., 2026]	Audio + text	Transformer	Appr. 2	0.367	0.579	0.63
BLM [Tong et al., 2026]	Text, video, image	Language	Appr. 2	0.47	0.53	0.89
		Vision		0.37	0.63	0.59
Protein PLM [Cheng et al., 2024]	Protein seq.	CLM	Appr. 2	0.578	0.421	1.37
		MLM		0.776	0.230	3.37
xLSTM [Beck et al., 2025]	Web text	Transformer	Appr. 2	0.575	0.424	1.36
		xLSTM		0.547	0.417	1.31
NMM [Shukor et al., 2025b]	Multimodal	Early fusion (avg.)	Appr. 3	0.526	0.473	1.11
		Late fusion		0.636	0.462	1.38
		Sparse early fusion		0.361	0.656	0.55
OLMo [Merrill et al., 2026]	Web text	Transformer	Appr. 3	0.46	0.54	0.85
		Hybrid		0.49	0.51	0.96
		Pure GDN		0.55	0.45	1.22
ProGen3 [Bhatnagar et al., 2025]	Protein seq.	ProGen3	Kaplan			1.48*
Dense Retrieval [Fang et al., 2024]	Query-doc pairs	Transformer	Kaplan			2.34*

* ProGen3 reports $N_{\text{opt}}(D) \propto D^{1.479}$ (Eq. 1); Dense Retrieval gives $\alpha = 0.56$, $\beta = 1.31$ (Eq. 10), yielding $\beta/\alpha \approx 2.34$.

† Exponents extracted from Figure S5 data.[7]

‡ Exponents originally from Kaplan et al. [Kaplan et al., 2020] (OpenWebText2) and Hoffmann et al. [Hoffmann et al., 2022] (MassiveText), as reported in DeepSeek.

Table A2: Published compute-allocation exponents from scaling law studies across domains. Columns a and b are the allocation exponents governing optimal model size ($N^* \propto C^a$) and dataset size ($D^* \propto C^b$). Ratio is a/b for Approach 2 and 3 rows; starred Kaplan-style rows use the quantities described above. Empirical a/b equals β/α only under the ideal Chinchilla model with $C \propto ND$. DeepSeek and DLM use non-embedding FLOPs per token as their model resource, rather than parameter count. Olmo Hybrid values are rounded free-fit estimates from Table 18; their 95% confidence intervals overlap, so the point estimates do not establish differences between architectures.

I Progressive Filtering for Chinchilla

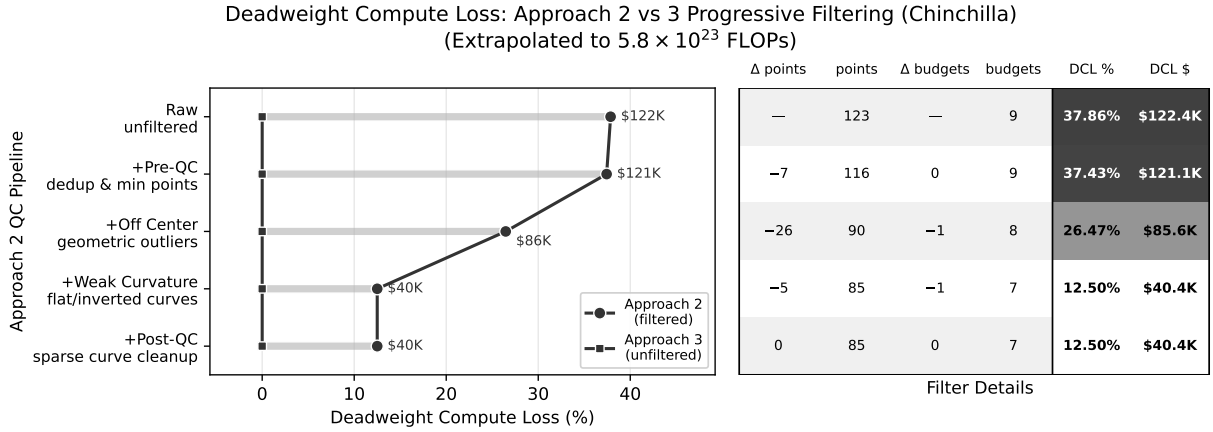


Figure A7: Progressive quality control filtering on Chinchilla data, analogous to Figure 2 for Llama 3. DCL is measured against an Approach 3 surface fit to unfiltered data and extrapolated to 5.8×10^{23} FLOPs. Starting from 37.9% DCL on raw data, the off-center and weak-curvature filters reduce DCL to 12.5%.

J Residual Distributions by Budget

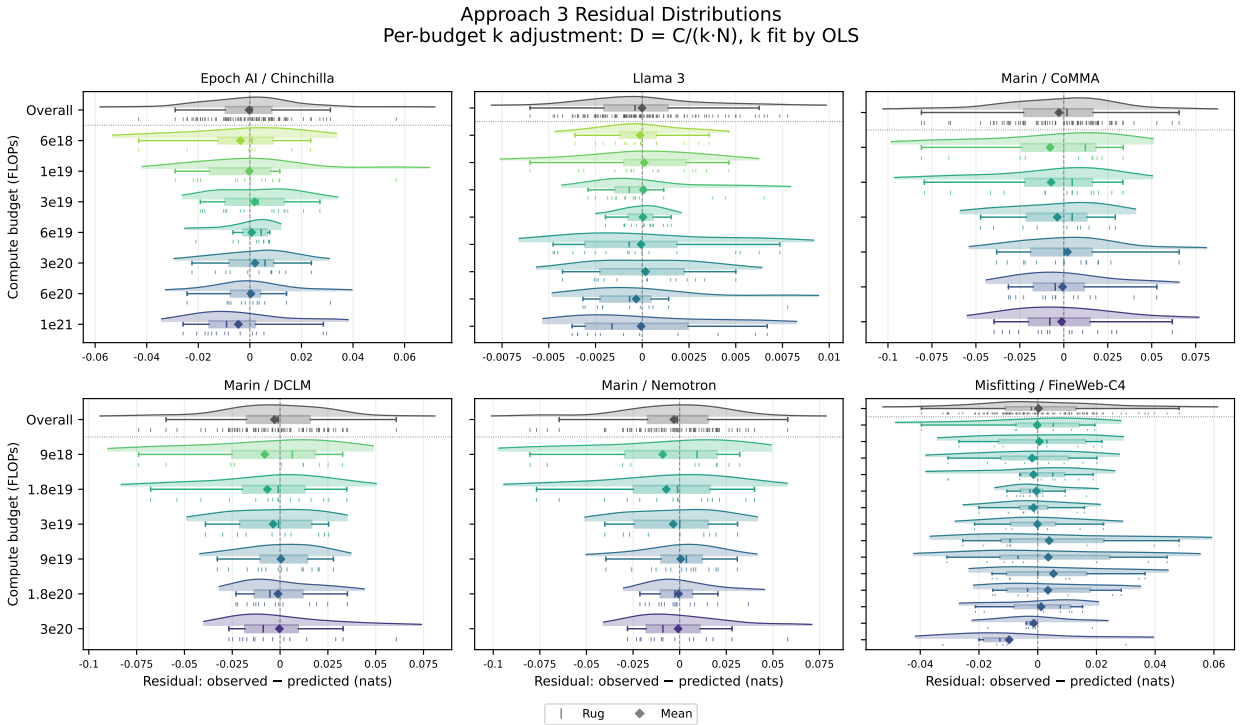
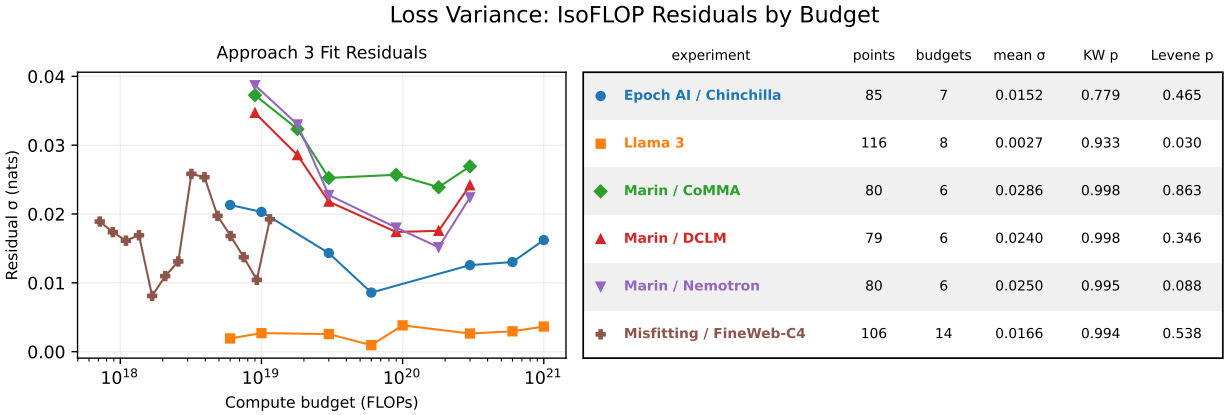


Figure A8: Per-budget residual distributions after Approach 3 fitting with per-budget FLOP factor adjustment, across six experiments. Each row shows the residuals for one compute budget as a layered rug plot, boxplot, and KDE, with an overall row pooling all clean points. The zero-residual reference line (dashed) indicates perfect prediction.

K Residual Variance Summary



experiment	points	budgets	mean σ	KW p	Levene p
Epoch AI / Chinchilla	85	7	0.0152	0.779	0.465
Llama 3	116	8	0.0027	0.933	0.030
Marin / CoMMA	80	6	0.0286	0.998	0.863
Marin / DCLM	79	6	0.0240	0.998	0.346
Marin / Nemotron	80	6	0.0250	0.995	0.088
Misfitting / FineWeb-C4	106	14	0.0166	0.994	0.538

Figure A9: Per-budget residual standard deviation (left) and summary statistics (right) for six experiments. Kruskal-Wallis tests whether residual locations differ across budgets. Levene tests whether residual variances differ across budgets. High p-values in most experiments indicate that a budget-independent noise model is a reasonable approximation for simulation purposes.

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